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Full Length Article

Pore-scale analysis of nanoparticle diffusion in brain tumours

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ABSTRACT

Nanoparticles have emerged as a promising platform for drug delivery to brain tumours. Despite their ability to successfully traverse the blood–brain barrier, nanoparticle penetration in tumour tissues, primarily governed by diffusion, remains significantly limited, posing a major challenge to effective delivery. The diffusion of nanoparticles in tumour tissues is determined by complex interactions between nanoparticles and the tumour microenvironment, a process that remains insufficiently understood. This study employs a mechanics-based model at the pore-scale to address this gap. After validation with reported experimental results, the model is applied to investigate nanoparticle diffusion across different grades of brain tumours under various conditions, with the 3D geometries of tumour microstructures mathematically reconstructed based on their morphological characteristics. The results indicate nanoparticles diffuse slowly in high-grade tumours despite their loose cell arrangements. This implies that the density of hyaluronic acid, the key tumour extracellular matrix component, surpasses tissue porosity in determining nanoparticle diffusion. To quantify nanoparticle diffusion, for each grade of brain tumours empirical formulas are developed to express the relationships between nanoparticle diffusion coefficient and hyaluronic acid concentration, as well as with nanoparticle size and zeta potential. Furthermore, the results demonstrate the Einstein-Stokes equation can be used to estimate nanoparticle diffusion coefficient at different temperatures by scaling the values at normal body temperature, whereas using it to directly calculate nanoparticle diffusivity would result in substantial errors. The developed model and empirical formulas provide effective tools for rapidly predicting nanoparticle diffusion, offering insights for drug nanocarriers' design to improve brain tumour treatment.

1. Introduction

Astrocytoma is a common type of brain tumour that originates from star-shaped astrocyte cells. High-grade astrocytoma is distinguished by its aggressive and malignant nature, which significantly contributes to its high mortality rate (Brodbelt et al., 2015; Smoll & Hamilton, 2014). Despite advancements in diagnosis and treatment modalities, the management of these brain tumours remains a formidable task, as conventional chemotherapies often fall short of achieving the desired outcomes in clinical practice (Koo et al., 2006). These routine treatment modalities often face challenges in effectively targeting these tumours while minimising the negative impact on healthy brain tissues (Owonikoko et al., 2014). Additionally, the intricacies of the brain microenvironment,

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including the blood-brain barrier (BBB) (Arvanitis et al., 2019) and heterogeneous characteristics (Jain et al., 2007), pose significant obstacles to effective and localised drug delivery.

In recent decades, the emergence of nanoparticles (NPs) has provided a promising avenue for improving the precision and efficacy of brain tumour therapy (Zhan, 2019, 2017; Zhan & Wang, 2018b, 2018a; Zhan & Xu, 2013). Specifically, NPs can be engineered with tailored formulations to improve their passage across the BBB, either by exploiting transport mechanisms such as receptor-mediated transcytosis or by disrupting the BBB temporarily to allow NPs passage (Chen et al., 2022; Kulkarni & Feng, 2013; Lockman et al., 2003, 2004; Qiao et al., 2012). Various NPs have since been developed with excellent anticancer efficacy presented in preclinical trials (Byeon et al., 2016; Hua et al., 2011; Jatyan et al., 2022; Koziara et al., 2004; Madhankumar et al., 2009). However, currently prevalent NPs still cannot fully meet clinical requirements. Subsequent analyses have shown that most NPs tend to accumulate passively around the microvasculature, failing to penetrate deeply into tumour tissue for effective therapy (Wong et al., 2011). This limited penetration is largely due to their poor diffusion in tumour tissues.

NPs diffuse within the tumour extracellular space (ECS), which is filled with interstitial fluid (ISF) and extracellular matrix (ECM), as depicted in the multi-scale schematic diagrams in Fig. 1. The ECM is a complex assembly of macromolecules, including proteins, glycosaminoglycans and proteoglycans. In contrast to normal brain tissues, the tumour ECM becomes denser (Lu et al., 2012; Mohiuddin & Wakimoto, 2021) due to the overexpression of various ECM components (Delpech et al., 1993; Novak & Kaye, 2000), particularly hyaluronic acid (HA). The intricate network of HA intertwined with proteoglycans and glycoproteins (Zimmermann & Dours-Zimmermann, 2008) has been identified as the significant barrier restricting NP diffusion within the ECS (Mohiuddin & Wakimoto, 2021). Moreover, the diffusion of NPs in tumour tissues is influenced by their intrinsic properties, such as NP size and zeta potential. For instance, compared to conventional 100 nm NPs, neutrally charged NPs capable of shrinking to 10 nm after extravasation from blood vessels demonstrated superior deep tissue penetration (Wong et al., 2011). Larger NPs with sizes up to 114 nm were able to diffuse to deep tissues when coated with a dense poly(ethylene glycol) layer, achieving a zeta potential of −5 mV (Nance et al., 2012). In practice, the factors that determine NP diffusion within a tumour are interrelated. The individual effects of each factor and their importance are poorly understood.

Studying NP diffusion in brain tumours through experimental means remains challenging. Methods, such as fluorescent staining (Tréhin et al., 2006), isotopic labelling (Meyers et al., 2013) and the combination of imaging tracer with MRI (Allard et al., 2009), are effective for measuring NP biodistribution at the tissue scale. However, they still lack the resolution necessary to accurately track the diffusion of each NP within the tumour ECS at the microscale. As an alternative, mathematical modelling approaches provide a promising solution. At the macroscale, which corresponds to entire brain tumours with characteristic dimensions on the order of centimetres, NP diffusion can be effectively described using a concentration field governed by a partial differential equation (Arifin et al., 2009; Levin et al., 1980; Siepmann et al., 2006). In this framework, the diffusion coefficient serves as a key parameter characterising the rate of NP transport within tumour tissue, influenced by both the properties of the NPs and the characteristics of the tumour microenvironment. At the pore-scale, referring to the scale at which microstructural features control the NP diffusion behaviour and can be identified with the characteristic size of the ECS between tumour cells. This typically spans several tens of micrometres, and NP diffusion is represented by the motion of individual particles. This is because the size of NPs, generally ranging from a few to several hundred nanometres, is comparable to the spatial dimensions of the tumour microenvironment. By employing a set of governing equations to describe the forces acting on individual NP, this approach captures the movement of each NP and enables tracking of their trajectories within tumour microstructures, which can be reconstructed from ECS morphological features extracted from microscopic images. Compared to the macroscale model, the pore-scale approach allows for a more detailed investigation of the

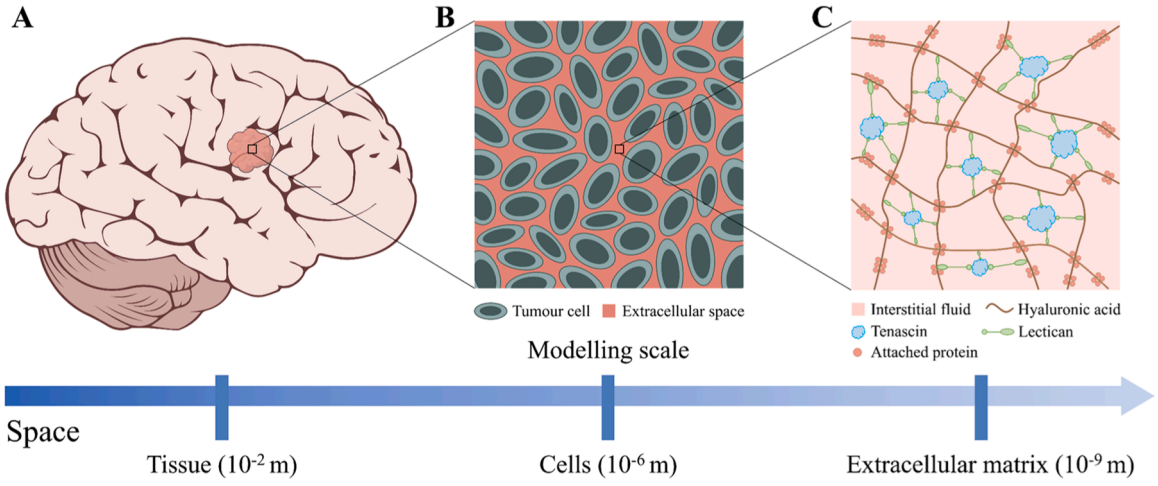


Fig. 1. Schematic diagrams of brain tumour tissue at different scales. (A) Macro/tissue scale: the brain tumour is within the brain. This image is adapted with permission from SciDraw.io (Smith Breault, 2020). (B) Microscale: the brain tumour is a porous medium, including the ECS (marked in red) and tumour cells (marked in green). The ISF can perfuse the entire tumour microstructures. (C) Nanoscale: ECS is occupied by the ISF and ECM primarily composed of the HA networks linked with other components, e.g. tenascin and lecticans. HA network is also attached to many proteins.

influence of individual factors. This facilitates a deeper understanding of the underlying mechanisms governing NP diffusion in brain tumours and provides valuable insights for optimising NP design to enhance therapeutic efficacy. This method has been effectively applied to study NP diffusion in the brain white matter (Yuan et al., 2022, 2023, 2024). Its application in the context of brain tumours is still unexplored.

This study aims to investigate NP diffusion in brain tumours and the impact of various influential factors. To tackle this challenge, we develop a pore-scale modelling framework that provides a high-resolution description of NP diffusion in the microstructures of brain tumours. This is achieved by integrating multiple functional modules, including an algorithm for reconstructing 3D brain tumour microstructures, a discrete model for tracking each NP movement based on Newton's 2nd Law, and a protocol for determining the diffusion coefficient, a macroscale parameter that directly quantifies the ability of NPs to diffuse in brain tumour tissues. After validating its accuracy, this framework is applied to assess the impact of various common factors representing NP and tumour properties. This approach employs mechanical modelling to investigate physical processes within biological systems. In particular, based on NP diffusion trajectories obtained at the pore-scale, we quantify NP diffusion by deriving their diffusion coefficients and formulating empirical relationships between this macroscale NP property and its underlying influencing factors.

2. Materials and methods

2.1. Mathematical model

Each NP interacts with other NPs, surrounding ISF and ECM, and cell membranes when travelling in the tumour ECS. To focus exclusively on the transport mechanism of diffusion, other mechanisms (Zhan et al., 2014b; Zhan, Gedroyc et al., 2017), such as convection which describes NPs travel with the ISF flow, drug release dynamics, and cellular uptake of NPs, are not considered. The motion of NPs in the tumour ECS in Fig. 1B is determined by the forces acting on them and spatially constrained by the surrounding cell membranes, as illustrated in Fig. 2.

NPs are assumed to be immersed in and travel through a homogeneous ECM. Thus, the hydrophilic or hydrophobic interactions between NP and the ECM are evenly distributed across the spherical surface of the NPs, resulting in mutual cancellation. The mathematical model is developed based on Newton's second law to describe the motion of an NP, as

$$m_p \frac{d\boldsymbol{u}_p}{dt} = \boldsymbol{F} = \boldsymbol{F}_B + \boldsymbol{F}_C + \boldsymbol{F}_D + \boldsymbol{F}_G + \boldsymbol{F}_{BY} + \boldsymbol{F}_{DLVO} \tag{1}$$

where m_p is the particle mass. $\boldsymbol{u}_p$ is the particle velocity. $\boldsymbol{F}$ is the overall net force acting on a particle. Specifically, $\boldsymbol{F}_B$ is the Brownian force. $\boldsymbol{F}_C$ is the Coulomb force due to the electrical interactions between particles. $\boldsymbol{F}_D$ is the drag force induced by the resistance of ISF and ECM. $\boldsymbol{F}_G$ is the gravity, and $\boldsymbol{F}_{BY}$ is the buoyancy force. As the cell membranes are negatively charged (De Pasquale et al., 2020), $\boldsymbol{F}_{DLVO}$ is the force that membranes of brain tumour cells exert on the NP. The force direction and sign of a single particle depend on the total force and change with the diffusion time.

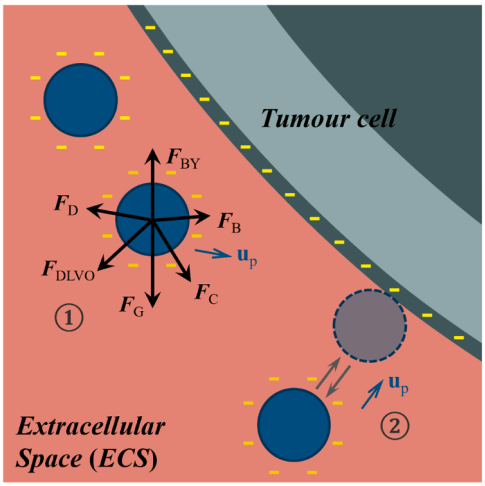


Fig. 2. Schematic diagram of the pore-scale model developed to simulate the motion of individual NPs within the tumour ECS. NPs, depicted as blue spheres, and the cell membrane are both negatively charged; these surface charges are indicated by yellow '-' signs. ① NP motion within the ECS is governed by the combined action of several forces, including gravitational force ($\boldsymbol{F}_G$), buoyancy force ($\boldsymbol{F}_{BY}$), Brownian motion ($\boldsymbol{F}_B$), drag force from ISF and the ECM ($\boldsymbol{F}_D$), Coulombic interactions between NPs ($\boldsymbol{F}_C$), and the force exerted by the cell membrane ($\boldsymbol{F}_{DLVO}$). $\mathbf{u}_p$ is the velocity of the NP. ② NPs are bounced back upon collision with the cell membrane. Please note that these two processes apply to all NPs within the ECS.

2.1.1. Brownian motion (Molecular thermal motion)

NPs exhibit Brownian motion when suspended in the tumour ECS. The Brownian force, $\boldsymbol{F}_{\mathrm{B}}$, is implemented as a Gaussian white noise process (Kim & Zydney, 2004). The mathematical expression can be given as

$$\boldsymbol{F}_{\mathrm{B}} = \chi\hat{\boldsymbol{e}}\sqrt{\frac{12\pi\mu_{\mathrm{eff}}k_{\mathrm{B}}Tr_{\mathrm{p}}}{\Delta t}} \tag{2}$$

where the χ denotes the Gaussian random number with zero mean and unit variance. $\hat{\boldsymbol{e}}$ is a unit vector with a random direction. μ_{eff} is the effective viscosity of the ISF. k_{B} is the Boltzmann constant. T is the absolute temperature with the unit K. r_{p} is the diameter of the NP, and Δt is the time step.

2.1.2. NP interactions (Coulomb force)

The NP surface is typically charged to prevent aggregation, as charged NPs repel each other when they get close, therefore enhancing NP diffusion. In the context of chemotherapy for brain tumours, NPs often possess a charge due to several factors, primarily the surface coatings (e.g. glucuronic acid) of the drug carriers and the biomolecules (e.g. nucleic acids and proteins) they interact with (Chertok et al., 2010; Masserini, 2013). Therefore, the Coulomb force, $\boldsymbol{F}_{\mathrm{C}}$, is introduced to describe the electrical interactions between the target NP and other NPs, written as

$$\boldsymbol{F}_{\mathrm{C}} = \frac{qq_{\mathrm{i}}}{4\pi\varepsilon_0}\sum_{i=1}^{N-1}\frac{\boldsymbol{r}-\boldsymbol{r}_{\mathrm{i}}}{|\boldsymbol{r}-\boldsymbol{r}_{\mathrm{i}}|^3} \tag{3}$$

where q is the charge quantity of the target NP and q_{i} is the charge quantity of one of the other NPs. ε_0 is the vacuum permittivity. N is the total number of NPs in the domain. $\boldsymbol{r}$ represents the position of the target NP, and $\boldsymbol{r}_{\mathrm{i}}$ is the position of one of the other NPs.

It should be noted that the charge quantity of an NP cannot be directly calculated because of its small size, complex surface coatings, surrounding medium, etc. (Ayala et al., 2013). Instead, zeta potential is commonly used to characterise NP surface charge (Honary & Zahir, 2013a, 2013b; Modena et al., 2019). As a measure of the electric potential at the surface of a particle in a liquid medium, zeta potential plays a pivotal role in influencing NP behaviour and motility in the tumour microstructure. More precisely, charged NPs can generate an electric double layer (EDL) (W Verwey, 1947) around themselves. A tightly formed inner layer consists of ions with opposite charges, while a loosely arranged outer layer contains both positive and negative ions. By solving the 1D Poisson-Boltzmann equation, the relationship between the surface charge density and surface electrostatic potential of a flat plate can be quantified by the Gouy-Chapman equation (McLaughlin, 1989). This equation is further developed to describe the relationship between the surface charge density of NP and its zeta potential (Ge & Wang, 2017; Yuan et al., 2022), as

$$\delta = \sqrt{8N_{\mathrm{A}}c\varepsilon k_{\mathrm{B}}T}\sinh\left(\frac{e\xi}{2k_{\mathrm{B}}T}\right) \tag{4}$$

where δ is the surface charge density of the NP. N_{A} is the Avogadro constant. c is the ion concentration. ε stands for the permittivity of the solution. ξ is the zeta potential of the NP.

The charge quantity of an NP can be calculated as the product of its surface charge density and surface area, as

$$q = A_{\mathrm{p}}\delta = 4\pi r_{\mathrm{p}}^2\delta \tag{5}$$

where A_{p} is the surface area of an NP. r_p is the radius of the NP in this study, and NP is assumed to be spherical.

2.1.3. Drag force

The viscous drag force, $\boldsymbol{F}_{\mathrm{D}}$, acting on an NP can be described by Stokes's law (Stokes, 1851). The correction factor, C_{slip}, is applied to account for the slip boundary effect at the interface between the nano-sized particles and their surrounding fluid, as

$$\boldsymbol{F}_{\mathrm{D}} = -6\pi\mu_{\mathrm{eff}}r_{\mathrm{p}}\Delta\boldsymbol{u}_{\mathrm{p}}\Big/C_{\mathrm{slip}} \tag{6}$$

where $\Delta\boldsymbol{u}_{\mathrm{p}}$ is the relative velocity of the NP to ISF. The correction coefficient of C_{slip} is defined as (Sharaf et al., 2019),

$$C_{\mathrm{slip}} = 1 + 2K_{\mathrm{n}}\left(1.257 + 0.4e^{-\frac{1.1}{2K_{\mathrm{n}}}}\right) \tag{7}$$

where K_{n} is the Knudsen number. C_{slip} becomes more significant when the NP size is less than 15 nm (Sharaf et al., 2019).

2.1.4. Gravity and buoyancy force

The gravity and buoyancy force of the target NP are respectively defined as

$$\boldsymbol{F}_{\mathrm{G}} = -m_{\mathrm{p}}g\hat{\boldsymbol{z}} \tag{8}$$

$$\boldsymbol{F}_{\mathrm{BY}} = m_{\mathrm{p}} g \frac{\rho_{\mathrm{f}}}{\rho_{\mathrm{p}}} \widehat{\boldsymbol{z}} \tag{9}$$

where g is the standard acceleration of gravity. ρ_{f} and ρ_{p} stand for the density of ISF and the NP, respectively. $\widehat{\boldsymbol{z}}$ is the unit vector in the z-direction.

2.1.5. Particle-cell membrane interactions

The electrical interactions between NPs and cell membranes can be described by the Derjaguin-Landau-Verwey-Overbeek (DLVO) theory (Gooch, 2010), which consists of the effects of the van der Waals interaction and double-layer force, which is given by

$$\boldsymbol{F}_{\mathrm{DLVO}} = -\frac{\partial}{\partial L} W(L) \widehat{\boldsymbol{n}} = -\frac{\partial}{\partial L} [W(L)_{\mathrm{A}} + W(L)_{\mathrm{R}}] \widehat{\boldsymbol{n}} \tag{10}$$

where $W(L)_{\mathrm{R}}$ is the potential rising from electrostatic interactions between the target NP and cell membrane, while $W(L)_{\mathrm{A}}$ corresponds to the potential generated by the van der Waals force between the NP and cell membrane. $\widehat{\boldsymbol{n}}$ is a unit vector normal to the cell membrane at the point closest to the NP (Gooch, 2010). Since the NP size (nm) is usually several orders lower than the size of tumour cells ($12 - 14$ μm (Abramczyk et al., 2020)), the tumour cell membranes can be assumed as a flat plate. Therefore, these two potentials between the spherical NPs and cell membrane can be calculated by

$$W(L)_{\mathrm{A}} = -A d_{\mathrm{p}} / 12L \tag{11}$$

$$W(L)_{\mathrm{R}} = \frac{64 k_{\mathrm{B}} T \rho_{\infty} \gamma^2}{\kappa} e^{-\kappa L} \tag{12}$$

$$\gamma = \tanh\left(\frac{z e \Psi_{\mathrm{c}}}{4 k_{\mathrm{B}} T}\right) \tag{13}$$

$$\kappa = \sqrt{\frac{\rho_{\infty} e^2 z^2}{\varepsilon_0 \varepsilon_{\mathrm{r}} k_{\mathrm{B}} T}} \tag{14}$$

where A is the Hamaker constant. d_{p} is the diameter of the NP. L is the distance between the NP and cell membrane. ρ_{∞} is the number density of ions in the ISF. γ is the reduced surface potential, which is a dimensionless form that scales the surface potential Ψ_{c} relative to thermal energy. Ψ_{c} is the potential on the membranes of tumour cells. $1/\kappa$ stands for the thickness of the diffused EDL, also known as the Debye screening length. z is the valency of the ion, and ε_{r} is relative static permittivity.

2.1.6. Diffusion coefficient

The diffusion coefficient of NPs (D) can be determined based on the mean square displacement (MSD) of the NPs, which refers to the mean distance between the positions of NPs in relation to their reference position against time. This parameter can be obtained using the trajectories of all the NPs obtained from the simulations (Yuan et al., 2022), as

$$MSD = \left\langle \left| \boldsymbol{R}^2 \right| \right\rangle = \left\langle \left| \boldsymbol{r}(t) - \boldsymbol{r}_0 \right|^2 \right\rangle = \frac{1}{N} \sum_{i=1}^{N} \left| \boldsymbol{r}_{\mathrm{i}}(t) - \boldsymbol{r}_{\mathrm{i}}(0) \right|^2 \tag{15}$$

$$D = \left\langle \left| \boldsymbol{R}^2 \right| \right\rangle / 2nt \tag{16}$$

where $\left\langle \left| \boldsymbol{R}^2 \right| \right\rangle$ is the absolute value of MSD. Vector $\boldsymbol{r}_{\mathrm{i}}(t)$ is the position of the i th NP at the time t. Vector $\boldsymbol{r}_{\mathrm{i}}(0)$ is the initial position of the i th NP. n denotes the n-dimensional diffusion and is set as 3 in this study.

Table 1
Model parameters of brain tumour in different grades.

Brain tumour grades	Porosity (%)	Cell size (μm)[a]
Grade 1 (pilocytic astrocytoma)	35 – 39 (Vargová et al., 2003; Zamecnik, 2005)	10 – 12 (Yang & Zhan, 2022)
Grade 2 (diffuse astrocytoma)	28 – 30 (Vargová et al., 2003; Zamecnik, 2005)	11 – 12 (Yang & Zhan, 2022)
Grade 3 (anaplastic astrocytoma)	43 – 45 (Vargová et al., 2003; Zamecnik, 2005)	11 – 13 (Louis et al., 2016)
Grade 4 (glioblastoma viable region)	43 – 45 (Vargová et al., 2003; Zamecnik, 2005)	12 – 14 (Abramczyk et al., 2020)
Grade 4 (glioblastoma necrotic region)	57 – 59 (Vargová et al., 2003; Zamecnik, 2005)	12 – 14 (Abramczyk et al., 2020)

[a] Cell size refers to the length of the minor axis. The ratio between the length of the major and minor axis is 2.2 – 2.5 (Abramczyk et al., 2020).

2.2. Model geometry

Astrocytoma is classified into Grades 1–4 by the WHO, including pilocytic astrocytoma (PA), diffuse astrocytoma (DA), anaplastic astrocytoma (AA) and glioblastoma (GBM) (Louis et al., 2021), ranging from low-grade to high-grade tumours. The geometric characteristics, including tissue porosity and cell size, exhibit variations across all grades due to their physiological and physico-chemical processes, as detailed in Table 1. Brain tumour tissue is typically characterised as isotropic at the macroscopic level because of the random distribution and orientation of tumour cells at the microscale (Roberts et al., 2020).

In order to reconstruct 3D microstructures of tumour ECS through mathematical means, tumour cells are simplified into ellipsoids according to the shape of their main body, as shown in Fig. 3A. We first employ the Monte Carlo method (Miodownik et al., 1999) to randomly seed dots in a 300 $\times$ 300 $\times$ 300 μm domain and let them grow in any two perpendicular directions to form the tumour cells. The orientation of each cell is dynamically and subtly adjusted during cell growth to eliminate or minimise the overlap between any two neighbouring cells. This process is stopped when both the cell size and tissue porosity reach their experimental ranges in Table 1, respectively, controlled by the Lubachevsky-Stillinger procedure (Lubachevsky & Stillinger, 1990). Tumour cells are then subtracted from the entire domain to obtain the tumour ECS. Finally, only the core region with a dimension of 200 $\times$ 200 $\times$ 200 μm is adopted for the following modelling studies. This is because tumour cells close to the six domain sides lose their directional randomness due to confinement to these sides during cell growth. The reconstruction process is briefly introduced in Fig. 3 using the GBM variable region as a representative. Readers are recommended to refer to our previous study (Yang, Yuan, Panaitescu et al., 2024) for more details. The tissue microstructures are reconstructed for brain tumours in all grades, including GBM necrotic region (NR), as shown in Fig. A1 in Appendix A.

2.3. Model parameters

The parameters utilised in the modelling studies are summarised in Table 2. Since this study is focused on the diffusion of NPs, the transport mechanism of convection that describes the influence of ISF flow on NP motion is excluded. Therefore, ISF is set to be still, and its flow velocity is zero. The justifications for selecting other key parameters are given below.

2.3.1. Effective viscosity

Viscosity is an intrinsic property of a fluid that refers to its resistance to deformation at a given rate. This property of ISF in tumours has been demonstrated to be affected by the distribution, arrangement, and hydrophilicity of the ECM components (Jain, 1987), particularly HA (Pluen et al., 2001), a key tumour ECM molecule whose concentration is at least 4-fold higher than that in normal brain tissues (Delpech et al., 1993). Therefore, the parameter of ISF effective viscosity is introduced to account for the impact of ECM components. Due to the dominating role of HA in the tumour ECM formation (Belousov et al., 2019; Delpech et al., 1993), the HA gel viscosity is applied as the effective viscosity.

The HA gels, formulated using unhydrolysed and acid-hydrolysed HA with varying molecular weights and concentrations, were perfused with isotonic saline solutions (0.2 M NaCl) to measure their viscosity at the room temperature of 25 °C (Lam, 1988). The results depicted in Fig. 4A demonstrate the crucial influence of HA concentration in determining the HA gel viscosity. It is worth noting that HA in tumours is purified without the procedure of acid hydrolysis (Margolis, 1967) and its molecular ranges widely from 1.0 $\times$

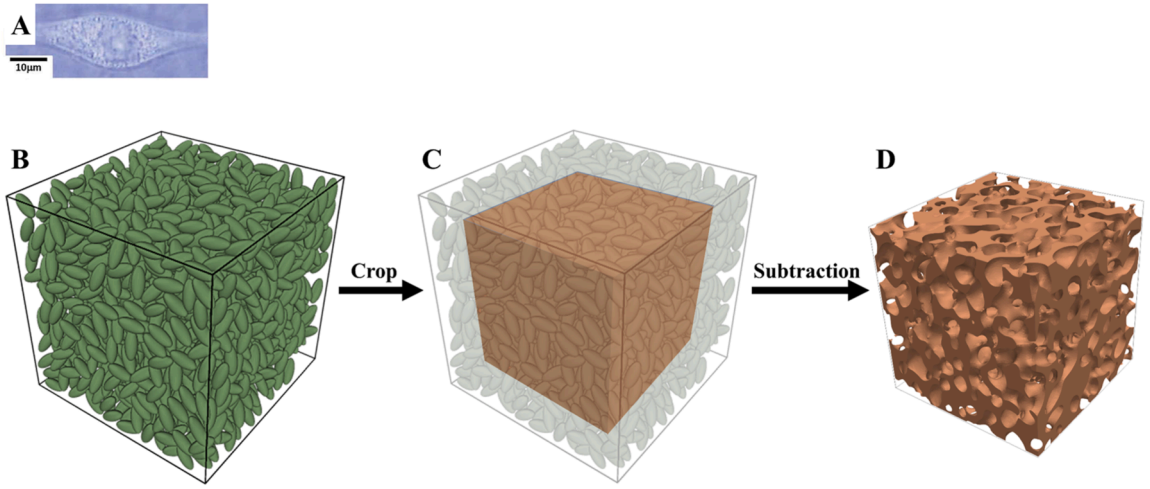


Fig. 3. Reconstruction of tumour 3D microstructure. (A) Microscopic image of a glioblastoma (U87) cell. Reprinted from Ref. (Abramczyk et al., 2020) under the Creative Commons Attribution-Share Alike 3.0 Unported license. (B) Generation of packed tumour cells. (C) The core region is subtracted and cropped to obtain the tumour 3D microstructure for modelling studies. (D) A representative reconstructed microstructure of the GBM viable region, utilised in the modelling analysis conducted in this study.

Table 2
Model parameters used in this study.

Symbol	Parameter	Unit	Value	Source
k_B	Boltzmann's constant	J/K	1.38×10^{-23}	(Sójka et al., 2006)
N_A	Avogadro constant	mol^{-1}	6.02×10^{23}	(Deslattes et al., 1974)
A	Hamaker constant	J	3.60×10^{-20}	(Faure et al., 2011)
K_n	Knudsen number	–	0.025	(Sharaf et al., 2019)
ρ_f	Density of ISF	kg/m^3	1.0×10^3	(Yang, Yuan, Rodriguez y Baena et al., 2024)
ρ_p	Density of NP	kg/m^3	2.0×10^3	(Kimoto et al., 2017)
ρ_∞	The number density of ions in the ISF	m^{-3}	8.73×10^{25}	(Leslie et al., 2019)
c	Ion concentration	mol/L	0.154	(Li et al., 2016)
e	Elementary charge	C	1.60×10^{-19}	(Hull & Williams, 1925)
ε	Permittivity of the solution	C/Vm	6.55×10^{-10}	(Gavish & Promislow, 2016)
z	Valency of the ion	–	1	(Hille et al., 1975)
ε_0	Vacuum permittivity	C/Vm	8.85×10^{-12}	(Urban et al., 2013)
ε_r	Relative static permittivity	–	80.1	(Archer & Wang, 1990)
Ψ_c	Zeta potential on the cell membrane	mV	-19.4 ± 1.2	(De Pasquale et al., 2020)

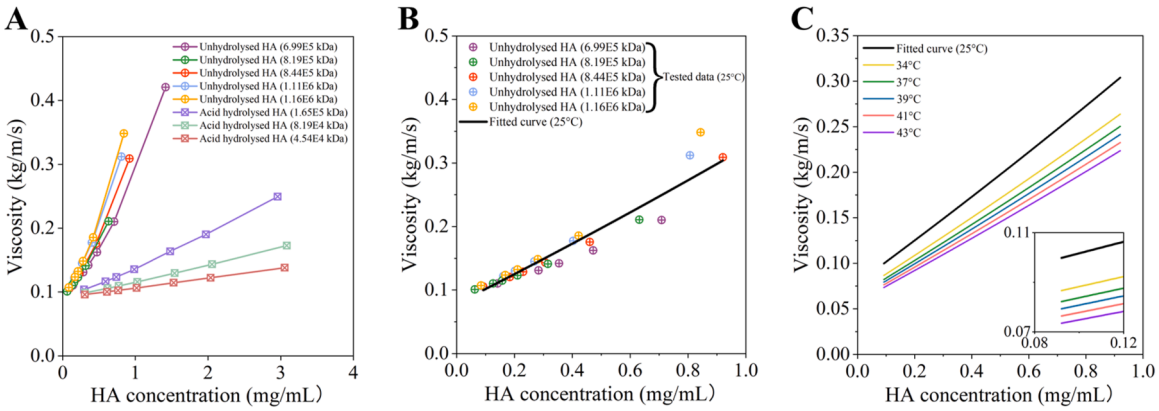


Fig. 4. Determination of the effective viscosity. (A) The viscosity of HA gel was measured across different HA concentrations and molecular weights, including unhydrolysed and acid-hydrolysed HA. Data is extracted from Ref. (Lam, 1988). (B) Derivation of the effective viscosity at room temperature. The adopted experimental data at 25 °C of unhydrolysed HA with different molecular weights are represented by coloured circles, while the fitted curve at 25 °C is shown as a black line with detailed fitting parameters. (C) Viscosity curves at different temperatures. They are obtained using Eq. (18) based on the fitted curve derived at 25 °C.

10^6 kDa to 1.0×10^7 kDa (Belousov et al., 2019; Novak & Kaye, 2000). Therefore, the experimental data from unhydrolysed HA with comparable molecular weights is selected to derive an empirical equation to correlate the HA gel viscosity at 25 °C and its concentration, as indicated by the fitted curve represented by the black line in Fig. 4B.

$$\mu_{HA}(C_{HA}, 25\ ^\circ C) = 1.69e^{\frac{C_{HA}}{7.33}} - 1.61 \tag{17}$$

where $\mu_{HA}(C_{HA}, 25\ ^\circ C)$ is the HA gel viscosity at 25 °C. C_{HA} is the HA concentration, with the unit in mg/mL. The temperature sensitivity of HA gel viscosity was measured experimentally (Rwei et al., 2008). By utilising the reported results (Rwei et al., 2008) to correlate HA gel viscosity with the environmental temperature, the viscosity of HA gel at varying temperatures and HA concentrations can be calculated by

Table 3
Effective viscosity of ISF of brain tumours in different grades.

Brain tumour grades	HA concentration range (mg/mL)	Effective viscosity range of ISF[a] (kg/m/s)
Grade 1 (pilocytic astrocytoma)	$3.3 \times 10^{-3} - 3.0 \times 10^{-1}$ (Delpech et al., 1993)	$6.53 \times 10^{-2} - 1.23 \times 10^{-1}$
Grade 2 (diffuse astrocytoma)	$3.3 \times 10^{-3} - 3.0 \times 10^{-1}$ (Delpech et al., 1993)	$6.53 \times 10^{-2} - 1.23 \times 10^{-1}$
Grade 3 (anaplastic astrocytoma)	$2.0 \times 10^{-2} - 8.1 \times 10^{-1}$ (Delpech et al., 1993)	$6.85 \times 10^{-2} - 2.27 \times 10^{-1}$
Grade 4 (glioblastoma viable region)	$5.1 \times 10^{-2} - 8.1 \times 10^{-1}$ (Delpech et al., 1993)	$7.44 \times 10^{-2} - 2.27 \times 10^{-1}$
Grade 4 (glioblastoma necrotic region)	$3.8 \times 10^{-2} - 6.1 \times 10^{-1}$ (Bertrand et al., 1992; Delpech et al., 1993)	$7.19 \times 10^{-2} - 1.85 \times 10^{-1}$

[a] ISF effective viscosity at 37 °C is calculated based on the fitted curve (green line) in Fig. 4C.

$$\mu_{HA}(C_{HA}, T) = [-0.0147(T - 273.15) + 1.367]\mu_{HA}(C_{HA}, 25\ ^{\circ}C) \tag{18}$$

where T is on the absolute temperature scale, with the unit in K. It can be converted to Celsius by $^{\circ}C = K - 273.15$. Specifically, the temperature and HA concentration are located in the ranges of $15 - 60\ ^{\circ}C$ and $0 - 3$ mg/mL, respectively.

The relationships between HA gel viscosity and HA concentration at representative temperatures ranging from 34 °C to 43 °C are presented in Fig. 4C. HA is considered homogeneously distributed in the tumour ECS microstructures since there is a lack of literature reporting its distribution and arrangement at the pore-scale. The ISF effective viscosity of brain tumours in different grades at the normal body temperature of 37 °C is calculated using their corresponding HA concentrations reported in the literature, summarised in Table 3.

2.3.2. Nanoparticle size

Various types of NPs have been designed for brain tumour therapy, such as doxorubicin-loaded NPs and paclitaxel-loaded NPs. Their sizes are typically located in the range from 10 nm to 200 nm (Ferraris et al., 2020; Hua et al., 2011; Jatyan et al., 2022; Koziara et al., 2004; Madhankumar et al., 2009). It is reported that 200 nm NPs with specific surface modifications also have the potential to cross the BBB to target the tumour site (Cheng et al., 2014; Lockman et al., 2002). Hence, the NPs size ranges from 10 nm to 200 nm in this study.

2.3.3. Zeta potential

NPs utilised in cancer therapy are usually negatively charged (Chertok et al., 2010; Masserini, 2013) to avoid NP aggregation and undesired interactions with negatively charged cell membranes and many proteins (Jo et al., 2015; Xiao et al., 2011). Compared to their positively charged counterparts, negatively charged NPs exhibit extended systemic circulation, enabling a more sustained drug

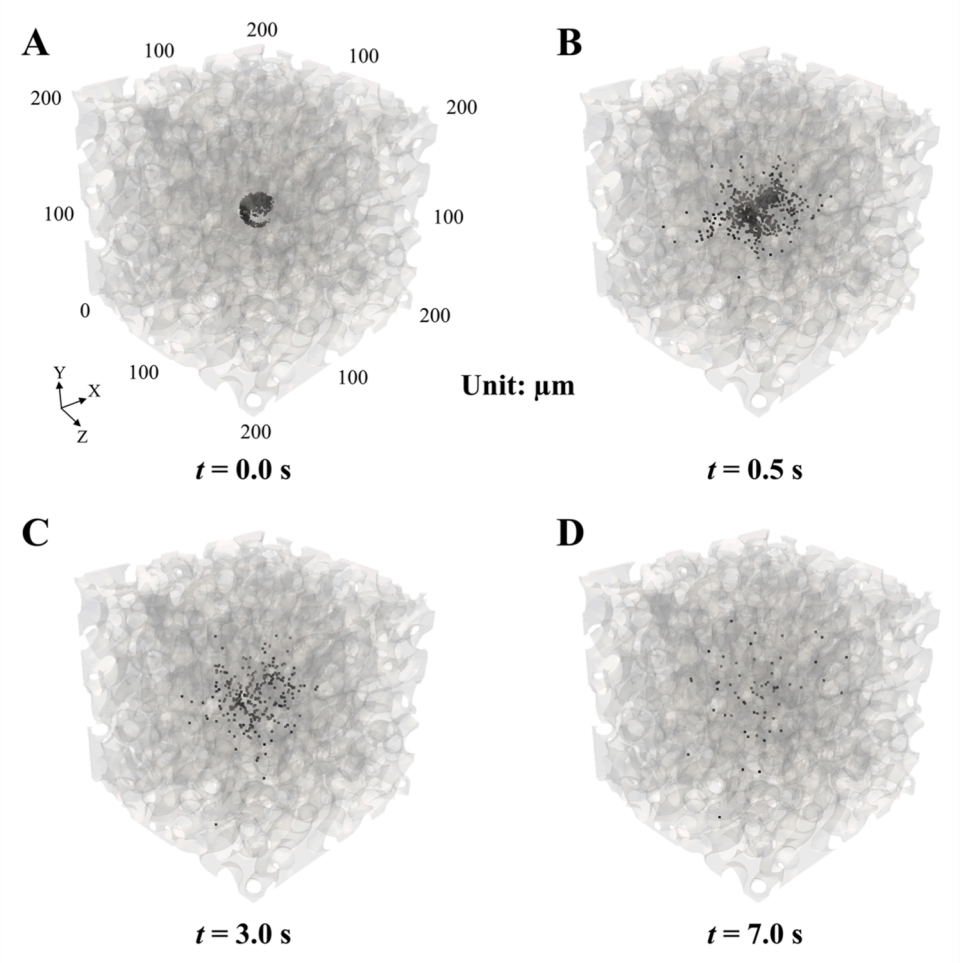


Fig. 5. Simulated NP distribution in the GBM viable region (Grade 4) under the experimental conditions that IgG diffuse in the xenograft model of human GBM (Pluen et al., 2001). (A)-(D) illustrate the NP dispersion patterns in tumour ECS microstructure at 0 s, 0.5 s, 3.0 s and 7.0 s, respectively.

supply (Chertok et al., 2010). Furthermore, experimental evidence highlights improved drug movement in matrix materials achieved by negatively charged NPs (Kim et al., 2010). The zeta potential varies significantly depending on the specific design of NPs, for instance, −2 mV (Nance et al., 2014), −12 mV (Byeon et al., 2016), and −44 mV to −32 mV (Chang et al., 2012). Therefore, zeta potential in the wide range from −50 mV to 0 mV is used in this study.

2.3.4. Tissue temperature

Tissue temperature is directly related to the Brownian motion of NPs and the effective viscosity. The reported normal brain temperature is located in the range from 36.1 °C to 40.9 °C (Rzechorzek et al., 2022). Besides, the temperature of the brain tumour can rise to 43 °C during local hyperthermia therapy (Skandalakis et al., 2020; Zhan et al., 2019) and may drop below 35 °C in extreme conditions, such as mild hypothermia (Enam et al., 2022). So, the range from 34 °C to 43 °C is employed for an exploratory study.

2.4. Boundary conditions

This model incorporates two boundary surfaces: the cell membranes and the outer surfaces of the tumour ECS microstructure domain. The cell membranes possess an electrical potential of −19.4 mV (De Pasquale et al., 2020), which repels NPs and thereby enables them to continue diffusing within the domain. NPs that collide with the cell membrane are reflected and continue to move within the domain. The mathematical formulation of this boundary condition is provided in Appendix B. This boundary condition allows us to focus on NP diffusion while disregarding other transport mechanisms, such as NP internalisation. NPs are excluded from the domain once reaching any of the domain's outer surfaces.

2.5. Model solving methods

An in-house Python program is developed to solve the governing equations in the mathematical model to simulate the diffusion of NPs. Model-solving methods for NP diffusion are described in detail in Appendix B. The workflow of the program built in Python for calculating NP diffusion in brain tumours is presented in Appendix C. 1000 NPs are initially released in a spherical volume at the centre of the tumour ECS microstructure domain to enable the diffusion process. This NP number is determined after a sensitivity test and is sufficient to obtain statistically reliable outcomes. Depending on the zeta potential of NPs in each simulation, the time step is located in the range from 10^{-10} s to 10^{-8} s, which is sufficiently fine to ensure statistically stable results after the time-step sensitivity tests.

3. Results

3.1. Model validation

The predictive accuracy of the present model should be validated prior to simulating NP diffusion. In the first validation study, the computed diffusion coefficient of IgG particle with a size of 12 nm (Pluen et al., 2001) and zeta potential of −6.5 mV (Yang et al., 2019) in viable GBM is compared with the experimental measurements obtained in the xenograft model of human GBM (Pluen et al., 2001). The average HA concentration in the GBM viable region is adopted. Shown in Fig. 5 are the simulated nano-sized particle dispersion in the GBM viable regions at different time points. The particles are initially packed in the centre of the domain. After being released at 0 s, they move towards the domain surfaces by diffusion in the tumour ECS, determined by the various forces outlined in

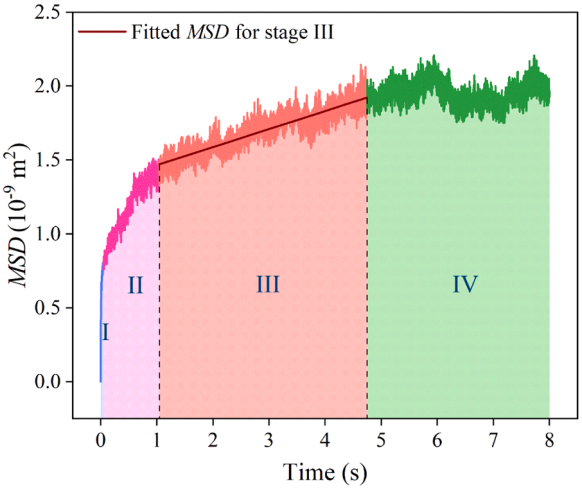


Fig. 6. The simulated time course of *MSD* for nano-sized IgG particles in viable GBM regions. The four stages are marked by different colours and Roman numbers. The curve fitting used to compute the NP diffusion coefficient is presented by the red line in Stage III.

Section 2.1. As a result, the distances between particles increase over time. Notably, most particles are still clustered in a small region at 0.5 s with short distances between them; while by 7.0 s, few particles remain in the domain as many have escaped after reaching the outer surfaces of the tumour ECS domain.

The simulated real-time locations and movement trajectories of all IgG particles over time are utilised to calculate the overall *MSD*, which is shown in Fig. 6 as a function of time. The entire process can be divided into four stages from I to IV. In Stage I, all particles are initially densely packed at the centre of the domain, as shown in Fig. 5A, and begin to diffuse into the tumour ECS upon release. This dynamic leads to a sudden increase in *MSD*. The persistent dispersion of NPs into the tumour ECS results in a gradual reduction in the *MSD* during Stage II, as shown in Fig. 5B. This progression continues until the slope of *MSD* becomes statistically stable. At this point, the system enters Stage III, where the particles are sufficiently spaced. This thereby provides a more accurate approximation of the actual process, in which particles, as a diluted substance, stably diffuse within tumour tissues, as indicated by Fig. 5C. However, as diffusion continues, an increasing number of particles escape from the domains, displayed in Fig. 5D, causing *MSD* to lose its statistical stability and exhibit significant fluctuations over time. This marks Stage IV. Therefore, the modelling results from Stage III are used to compute the diffusion coefficient via Eq. (16). The result is 2.03×10^{-11} m^2/s, which falls in the reported experimental range of $1.26 \times 10^{-11} - 2.54 \times 10^{-11}$ m^2/s (Pluen et al., 2001). Moreover, animal experiments also reported similar *MSD* trends of nano-sized tubes in the brain (Paviolo et al., 2020).

In the second validation study, the model is applied to compute the diffusion coefficient of uncharged NPs with a size of 2.95 nm, 14.1 nm and 35.4 nm in dilute agarose (0.3 %) gel at 37 °C (Thorne & Nicholson, 2006), respectively. The viscosity of the transport medium is set to 6.915×10^{-4} kg/m/s, extracted from Ref. (Thorne & Nicholson, 2006). The time courses of the overall *MSD* of these NPs are shown in Fig. 7. The results in Stage III, as highlighted in the zoom-in regions, are used to derive their diffusion coefficients. The comparisons with reported experimental results are summarised in Table 4, showing good agreements between the modelling and experimental results. These findings demonstrate the prediction accuracy of the developed model in this study.

3.2. Nanoparticle diffusion in tumours across grades

ECS, the medium where NP diffuses, differs between different grades of brain tumours due to the variations in tumour cell size, tissue porosity and ECM density. Modelling is performed to simulate NP diffusion in brain tumours from Grade 1 to Grade 4 including the necrotic regions of GBM; the HA concentration, which reflects the tumour ECM density, varies in the grade-specific ranges, respectively, as detailed in Table 3. The NP size, zeta potential and tissue temperature are maintained at -30 mV, 100 nm and 37 °C, respectively, to isolate the impact of tumour grade on NP diffusion.

Fig. 8A–E plots the representative time-dependent variations in the *MSD* of NPs within brain tumours of different grades, where the HA concentration is the average of their respective ranges. The similar trends observed in these time courses suggest that NPs undergo comparable movement stages, from Stage I to Stage IV, among tumour grades. However, the differences in amplitude and time scale across grades are subject to the combined effect of multiple tumour properties, including tissue porosity and HA concentration. The instantaneous views of NP dispersion at the midpoint of Stage III, shown in the insets, demonstrate that NPs are spreading into the tumour tissue at this stage. The substantial presence of NPs within the tumour ECS ensures statistically stable results, enabling a macroscopic description of different diffusion using NP diffusion coefficients.

The predicted time courses of *MSD* during Stage III in different grades of brain tumours are utilised to compute the NP diffusion coefficient, with the results presented in Fig. 9. The mean values of the diffusion coefficient are determined to be 1.69×10^{-9} m^2/s for PA (Grade 1), 1.47×10^{-9} m^2/s for DA (Grade 2), 1.33×10^{-9} m^2/s for AA (Grade 3), 1.26×10^{-9} m^2/s for the viable region of GBM (Grade 4), and 1.76×10^{-9} m^2/s for necrotic region of GBM (Grade 4). These findings indicate that, except in the necrotic regions of GBM, NP diffusion slows down as tumour grade increases. This trend in viable tumours is primarily due to tumour properties. Specifically, because of the comparable HA concentrations in Grade 1 and Grade 2 tumours, the higher diffusion coefficient in

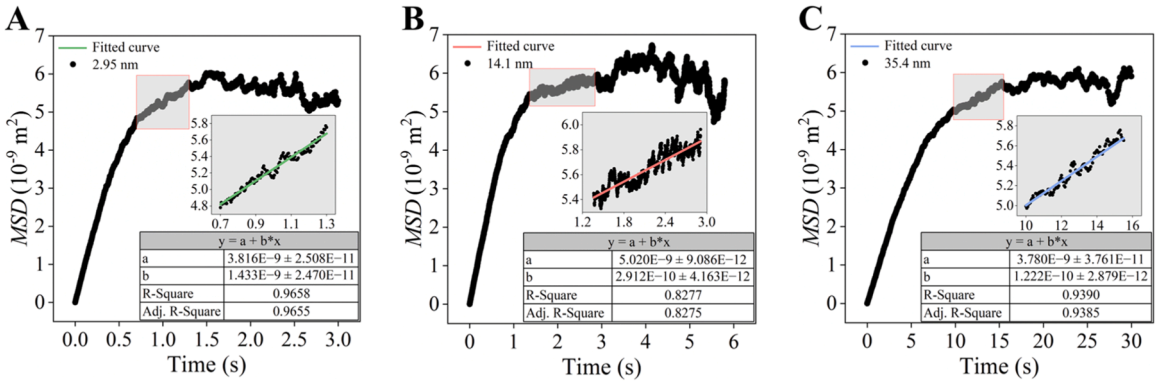


Fig. 7. The time courses of *MSD* of uncharged NPs with various sizes in dilute (0.3 %) agarose. (A) 2.95 nm NPs, (B) 14.1 nm NPs, (C) 35.4 nm NPs. *MSD* during Stage III, which are highlighted in the shadow, are applied to compute the NP diffusion coefficients. The fitted curve parameters are given in the inserted tables.

Table 4
Comparison between the experimentally measured and modelling predicted diffusion coefficients of NPs with different sizes in dilute (0.3 %) agarose.

Particle diameter d_p, nm	D (Experiment (Thorne & Nicholson, 2006)), 10^{-11} m^2/s	D (Present study), 10^{-11} m^2/s
2.95 (TR-dex3)	22.2 ± 0.16	23.9 ± 0.41
14.1 (TR-dex70)	4.67 ± 0.061	4.85 ± 0.069
35.4 (QD655)	1.86 ± 0.049	2.04 ± 0.048

Grade 1 tumour can be attributed to its higher tissue porosity, which suggests a larger ECS and less hindrance to NP diffusion. The slower diffusion in high-grade tumours results from the combined effect of tissue porosity and HA concentration. Although the higher tissue porosity of Grade 3 and Grade 4 tumours could facilitate faster NP passage, the elevated HA concentration further hinders NP movement by inducing a more viscous microenvironment with greater resistance. This finding also indicates that HA concentration has a more significant influence on NP diffusion than tissue porosity in brain tumours. Furthermore, since the tissue porosities are comparable in AA and the GBM viable region, the lower NP diffusion coefficient in the latter can be attributed to its slightly higher HA concentration. The results also show that NP diffusion is fastest in the necrotic region of GBM, due to the joint effect of its lower HA concentration and highest tissue porosity.

3.3. Effect of HA concentration

The HA concentration can vary significantly among different grades of brain tumours, as indicated by the ranges shown in Table 3. Parametric studies are conducted to assess its impact on NP diffusion in each grade of brain tumour. The artificial scenario of zero HA concentration is applied as a control group to elucidate and distinguish the effect of HA on the diffusion behaviours of NPs. Fig. 10A–E present the computed NP diffusion coefficient as a function of HA concentration in PA, DA, AA, GBM viable region and necrotic region, respectively. The results demonstrate that HA concentration plays a critical role in NP diffusion. Comparisons to the control group indicate that neglecting HA concentration leads to a several-fold overestimation of NP diffusion coefficient across all tumour grades. This discrepancy arises from the resistance generated by HA, which impacts NP diffusion through its influence on the Brownian motion and ISF drag force, as defined by Eqs. (2) and (6), respectively. Moreover, the results reveal that the NP diffusion coefficient exhibits an exponential decrease with increasing HA concentration in brain tumours of all grades. Empirical equations are derived to quantitatively describe this relationship, facilitating rapid estimation of NP diffusion coefficient in a tumour grade-specific manner. The curve-fitting results are shown in Fig. 10A–E, with the corresponding equations summarised in Table 5.

3.4. Effect of zeta potential

Zeta potential is a commonly used parameter to assess the surface charge of NPs. It can vary considerably depending on the specific NP formulation. The effect of zeta potential on NP diffusion is examined using the GBM viable region model, selected as a representative brain tumour due to its status as the most aggressive and lethal among the four grades. The tissue temperature is 37 °C and the average HA concentration in the GBM viable region is adopted. The results, presented in Fig. 11A, show that the diffusion coefficient increases exponentially with the magnitude of zeta potential. This behaviour is attributed to the positive relationship described by Eqs. (4) and (5), which demonstrate that a higher negative zeta potential corresponds to an increased charge quantity, as depicted in Fig. 11B. The elevated charge can effectively enhance the Coulomb force, which is the repulsion between NPs, thereby accelerating their diffusion. An interesting finding in the zoomed-in insert shows that for neutral NPs, larger NP sizes result in lower diffusion coefficients. This is primarily due to the slower Brownian motion and higher drag force, as described by Eqs. (2) and (6), respectively. However, larger NPs present faster diffusion when they carry a charge, e.g. -10 mV. This suggests that, in charged NPs, the Coulomb force becomes the dominant factor in determining NP diffusion, outweighing the influence of Brownian motion and drag force.

3.5. Effect of nanoparticle size

Fig. 12A illustrates the computed diffusion coefficient of NPs with varying zeta potentials as a function of their size within the GBM viable region at 37 °C. For negatively charged NPs, an exponential increase in the diffusion coefficient with NP size is observed. This seemly counterintuitive finding can be attributed to the physical meaning of zeta potential, which reflects the surface charge density (δ), as demonstrated by Eq. (4), rather than the total charge an NP has. Since the surface charge density remains the same when zeta potential is fixed, larger NPs with greater surface areas have higher charge quantity, which is shown logarithmically in Fig. 11B This enhances the Coulomb force, thereby accelerating NP diffusion in tumour ECS, as denoted by Eq. (3).

To further substantiate this explication, an additional parametric study is performed to simulate the diffusion of NPs with a fixed charge quantity of $-100\ e$, which is artificially assigned. Uncharged NPs are used as the control group for comparison. The computed diffusion coefficients of these two NPs are plotted against their sizes in Fig. 12B. It is not surprising that both NPs diffuse slower when their sizes become larger. This is because, for the NPs with fixed charge quantity, the increase in NP size only results in diminished thermal motion and augmented ISF drag force, both of which impede the NP movement.

The results presented in Figs. 11 and 12 suggest a strong interconnection between NP size and zeta potential. To further quantify their contributions to the determination of the NP diffusion coefficient, an empirical equation has been formulated using the modelling results, as

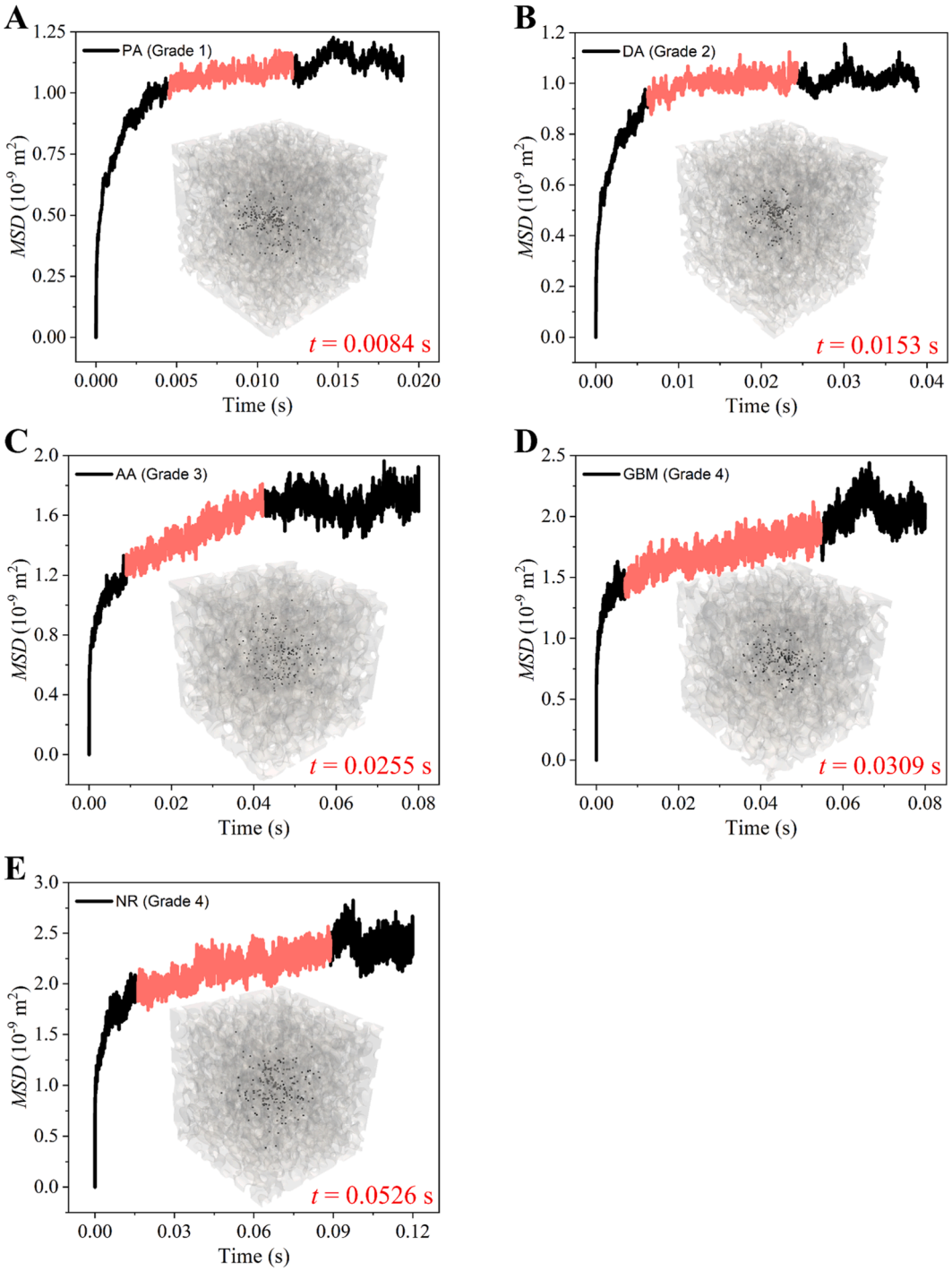


Fig. 8. Comparison of NP diffusion in the ECS microstructure of brain tumours in different grades. (A)-(E) Time course of *MSD* and representative NP dispersion patterns in PA, DA, AA, GBM viable region and necrotic region, respectively. All the dispersion plots are selected at the midpoint of stage III. The average HA concentration in each tumour grade is adopted.

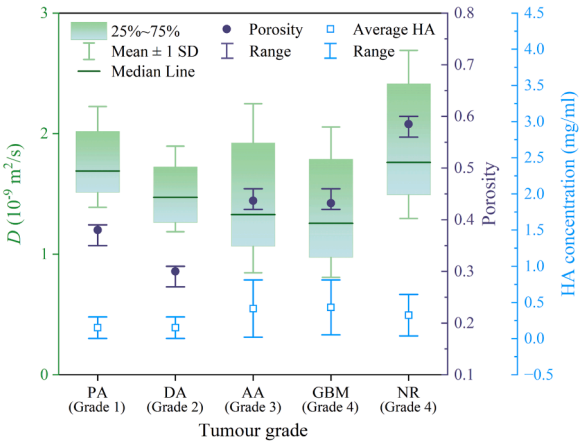


Fig. 9. Comparison of diffusion coefficient across different grades of brain tumours. The detailed diffusion coefficient is presented in a box plot on the left y-axis, while the porosity and HA concentration range of brain tumours in different grades are referenced to the right y-axis, marked in aubergine and blue, respectively.

$$D = 10.71 \times \frac{\exp\left(1.69 \times 10^{-6}\xi^3 - 7.70 \times 10^{-2}\xi\right)}{\exp\left(1.29 \times 10^{-7}d_p^3 - 2.60 \times 10^{-2}d_p\right)} \quad \left(-50\text{mV} \le \xi \le -10\text{mV}, 10\text{nm} \le d_p \le 200\text{nm}\right) \tag{19}$$

where D is the diffusion coefficient of NPs ($\mu m^2/s$). d_p and ξ are the particle size (nm) and zeta potential (mV), respectively. The result of surface fitting to the computed NP diffusion coefficient is displayed in Fig. 13, where the R^2 (coefficient of determination) and adjusted R^2 of this empirical equation are 0.9990 and 0.9987, respectively.

3.6. Effect of tissue temperature

The diffusion coefficient of NPs is also examined in relation to tissue temperature. The NP size is maintained at 100 nm, with the zeta potential set to be −30 mV and 0 mV, respectively. The average HA concentration in the GBM viable region is adopted. The results in Fig. 14 show that the NP diffusion coefficient is positively correlated to tissue temperature. This is because higher temperatures, on the one hand, can effectively enhance the Brownian motion of NPs; on the other hand, reduce the effective viscosity of ISF and thereby lower the ISF drag force and resistance to NP diffusion.

The computed NP diffusion coefficients are further compared to the values estimated using the Stokes-Einstein equation, as

$$D = \frac{k_B T}{3\pi\mu_{eff} d_p} \tag{20}$$

Fig. 14 reveals that the Stokes-Einstein equation overestimates the diffusion coefficient of uncharged NPs. This occurs as the equation is developed to describe the diffusion of spherical particles through a single pure liquid region, which means that the resistance introduced by the tumour ECS microstructure, which behaves as a porous medium, is not taken into account. In contrast, the comparison also shows that the diffusion coefficient of charged NPs can be underestimated by the Stokes-Einstein equation since it cannot account for the repulsion of Coulomb forces between NPs.

An analysis is carried out to explore the possibility of scaling the NP diffusion coefficient at 37 °C using the Stokes-Einstein equation to obtain values at other temperatures, as

$$D(T) = D_{37\ ^\circ C}\left[\frac{k_B T}{3\pi\mu_{eff}(T)d_p} \Big/ \frac{310.15k_B}{3\pi\mu_{eff,37\ ^\circ C}d_p}\right] \tag{21}$$

where $D(T)$ is the NP diffusion coefficient at the target temperature, with the unit of K. $D_{37\ ^\circ C}$ is the NP diffusion at the normal body temperature of 37 °C. Similarly, $\mu_{eff,37\ ^\circ C}$ and $\mu_{eff}(T)$ are the ISF effective viscosity at 37 °C and the target temperature, respectively. The diffusion coefficients estimated through scaling are compared with those computed from the current model in Fig. 15. The results indicate that scaling with the Stokes-Einstein equation slightly underestimates NP diffusion coefficients at the tissue temperature ranging from 34 °C to 43 °C. Nevertheless, the relative differences remain smaller than 2 %, demonstrating that this scaling approach is a feasible method for estimating diffusion coefficients at other temperatures in the examined temperature range.

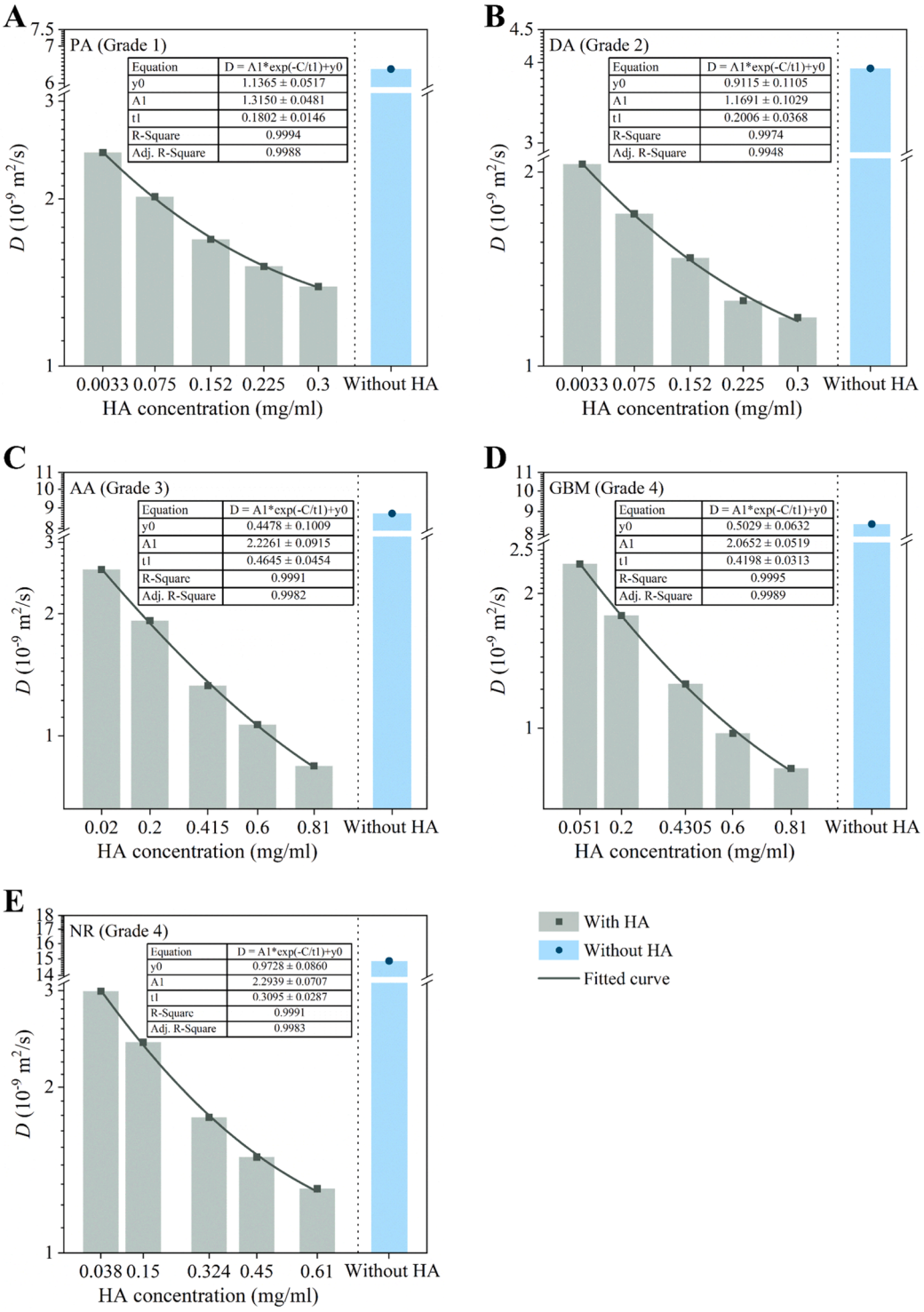


Fig. 10. Comparison of NP diffusion in tumours across different grades with and without HA. (A)-(E) The relationship between the HA concentration and diffusion coefficient in PA, DA, AA, GBM viable region and necrotic region, respectively. Curve-fitting results are utilised to derive the empirical equations for the NP diffusion coefficient as a function of HA concentration in each grade. The viscosity of ISF is set to be 6.9 $\times 10^{-4}$ kg /m/s (Goiriena-Goikoetxea et al., 2020) when excluding the impact of HA. The NP size, zeta potential and tissue temperature are fixed at 100 nm, −30 mV and 37 °C, respectively. The units of $A1$ and $y0$ are 10^{-9} m^2/s, while the unit of $t1$ is mg/ml.

Table 5
Empirical equations for NP diffusion coefficient (10^{-9} m^2/s) in brain tumours of different grades[a].

Brain tumour grade	Empirical equation
PA (Grade 1)	$D = 1.3150e^{-C_{HA}/0.1802} + 1.1365$
DA (Grade 2)	$D = 1.1691e^{-C_{HA}/0.2006} + 0.9115$
AA (Grade 3)	$D = 2.2261e^{-C_{HA}/0.4645} + 0.4478$
GBM viable region (Grade 4)	$D = 2.0652e^{-C_{HA}/0.4198} + 0.5029$
GBM necrotic region (Grade 4)	$D = 2.2939e^{-C_{HA}/0.3095} + 0.9728$

[a] The diffusion coefficient is developed for 100nm NP at 37 °C.

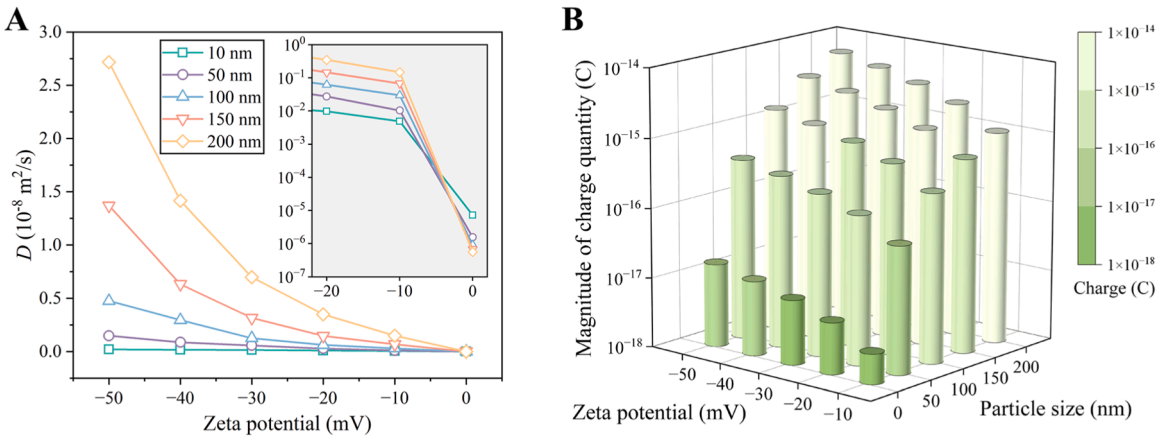


Fig. 11. The relationship between the zeta potential, diffusion coefficient, and absolute effective charge of NP. (A) Diffusion coefficient of NPs. (B) The magnitude of charge quantity of NP.

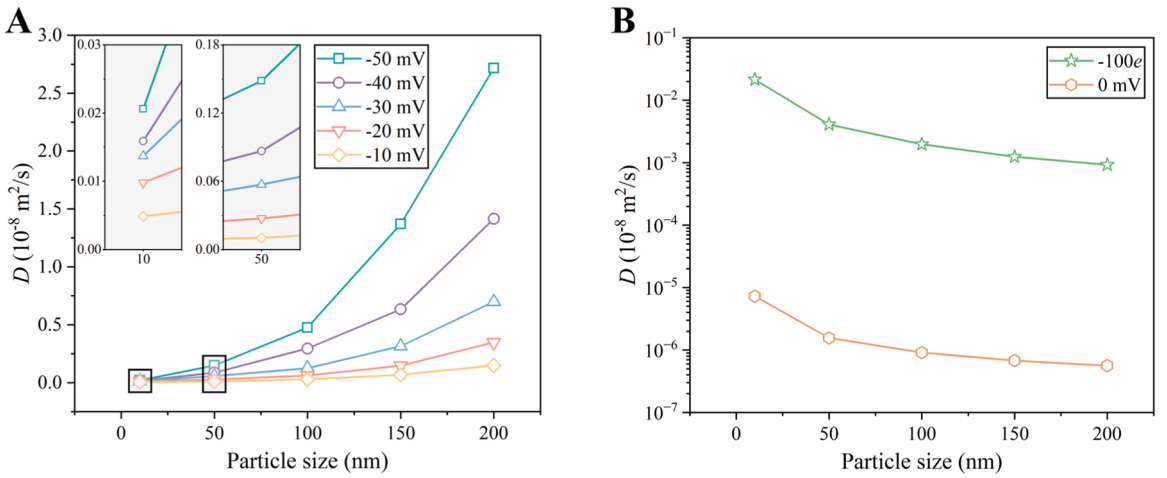


Fig. 12. The NP diffusion coefficient as a function of particle size. (A) NPs with different zeta potentials. (B) NPs with fixed absolute effective charge.

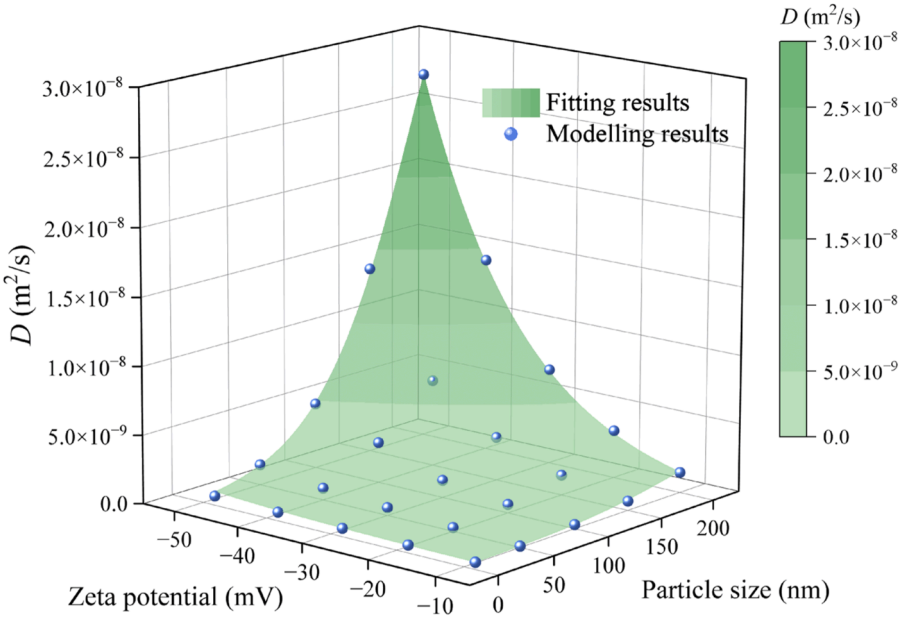


Fig. 13. Surface fitting results to derive the empirical equation for NP diffusion coefficient as a function of NP size and zeta potential.

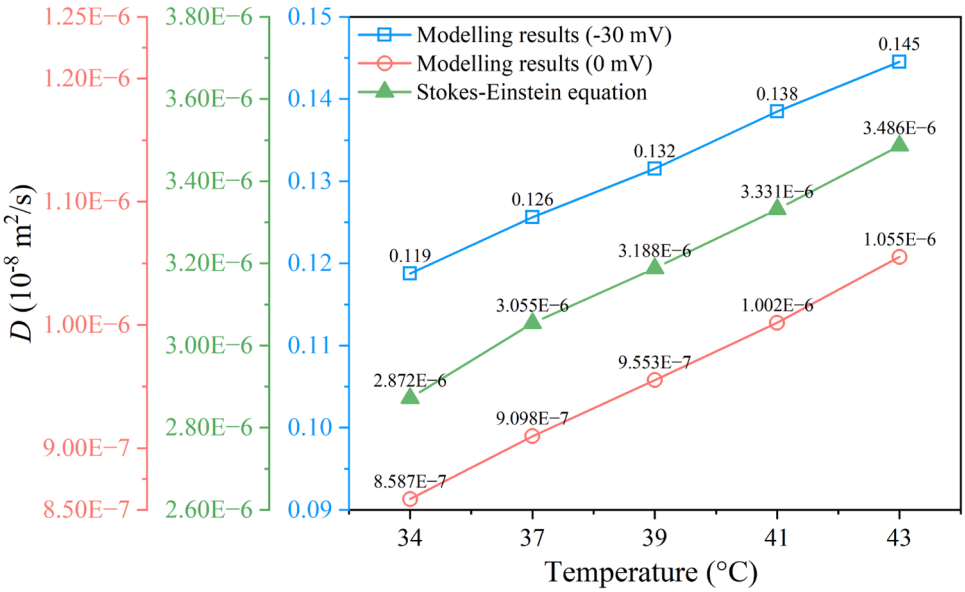


Fig. 14. Impact of tissue temperature on NP diffusion coefficient. The computed NP diffusion coefficients by the current model and the Stokes–Einstein equation at different tissue temperatures.

4. Discussion

The diffusion of NPs in brain tumours is determined by multiple factors, including tissue properties such as ECS microstructure, ECM density and temperature, as well as NP properties like particle size and zeta potential if the NPs carry a charge. Our modelling results highlight the critical role of NP surface charge, which can markedly enhance repulsive interactions between NPs and thereby accelerate their diffusion. For instance, 100 nm NPs with a zeta potential of −10 mV exhibit a diffusion coefficient that is 1500-fold higher than that of the neutral NPs of the same size, as illustrated in Fig. 12. This dramatic enhancement can be achieved by either increasing the zeta potential or the size of charged NPs. Given that NPs are commonly designed with a size close to 100 nm to minimise rapid elimination by the liver and spleen for prolonged circulation in the bloodstream (Liu et al., 1992), enhancing the zeta potential emerges as a more practical strategy. The potential of this approach is further supported by the exponential increase relationship

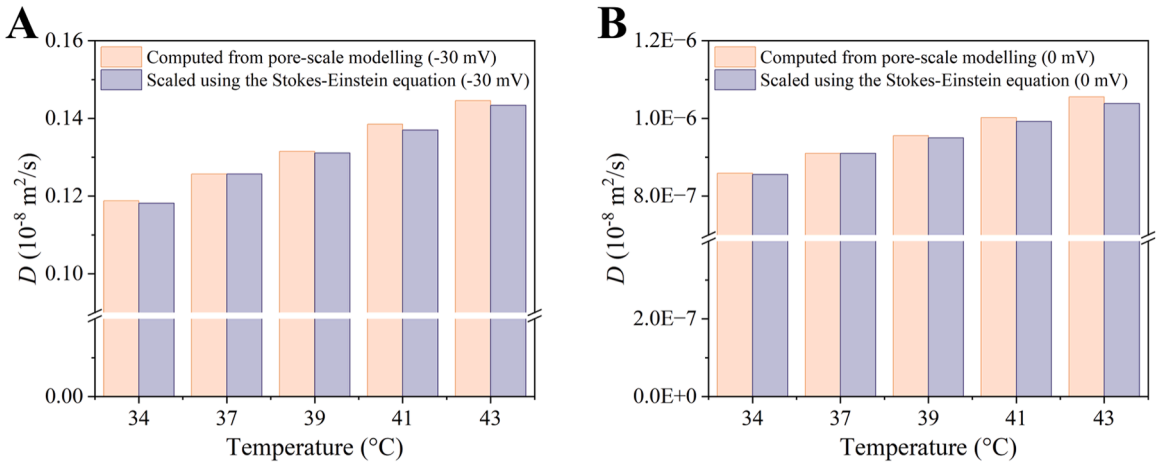


Fig. 15. Comparisons of diffusion coefficients obtained from the current model and estimated through scaling using the Stokes–Einstein equation and the diffusion coefficient at 37 °C. (A) Zeta potential is set to be − 30 mV. (B) Zeta potential is set to be 0 mV.

between zeta potential and NP diffusion coefficient, as demonstrated in Fig. 11. Enhancing zeta potential can be accomplished through surface modification using different biochemicals and NP fabrication techniques, providing a promising means to adjust NP behaviour for improved therapeutic delivery in brain tumours.

ECM density is shown to exceed tissue porosity in influencing NP diffusion across different grades of brain tumours. Specifically, NPs present a 5.54-fold lower diffusion coefficient in high-grade tumours even though they possess 33.3 % higher tissue porosity compared to low-grade tumours. This reduced diffusion is attributed to the dense ECM, particularly overexpression of HA, which is evaluated to be fourfold or more in high-grade brain tumours (Belousov et al., 2019; Delpech et al., 1993). Previous modelling studies on normal tissues usually neglect the influence of ECM on ISF viscosity and substance transport by assuming the ECS microstructure as free-transport channels (Holter et al., 2017; Yuan et al., 2025). However, findings from this study indicate that this assumption does not hold in brain tumours, where the ECM is essential in hindering NP diffusion. Furthermore, recent studies reported that ECM components such as HA are intimately involved in tumour cell proliferation and migration (Akiyama et al., 2001; Bellail et al., 2004). Advanced medications including NPs are designed to target and normalise tumour ECM (Kang et al., 2016; Raavé et al., 2018; van der Steen et al., 2017; Zhang et al., 2014). The reduced concentration of ECM components, in turn, may facilitate a more favourable environment for NPs to diffuse deeper into tumoural tissues to achieve a more homogeneous distribution. It is then reasonable to hypothesise that the delivery outcomes could be improved through a sequential delivery strategy involving ECM-targeting NPs followed by cytotoxic drug-loaded NPs. Nevertheless, further validation through *in vitro* and *in vivo* studies is essential to confirm its effectiveness.

Tissue temperature can notably influence NP diffusion, especially in localised chemotherapy using thermosensitive NPs under mild hyperthermia. Elevated tissue temperature can enhance NP diffusion by increasing their Brownian motion and reducing ECM impedance through the thinning of ISF effective viscosity. However, directly establishing the NP diffusion using the Stokes-Einstein equation can lead to significant inaccuracies. This is primarily because the equation does not account for the impact of complex tumour ECS microstructure and NP surface charge. Despite these limitations, the Stokes-Einstein equation can still be used to scale the NP diffusion coefficient at a specific temperature, e.g. 37 °C, providing a promising estimation of the diffusion coefficients at other temperatures, as shown in Fig. 15.

The particle tracing model is employed to simulate NP diffusion in 3D tumour microstructures. The resulting pore-scale trajectories of individual NPs are used to derive effective diffusion coefficients at the macroscale. The overall modelling strategy is outlined in Appendix D. Particularly, the inherent randomness of NP motion is captured in the model through three principal mechanisms. First, Brownian motion is explicitly incorporated, reflecting thermally induced random behaviour dependent on particle size and environmental temperature. Second, inter-particle interactions introduce further variability, as each NP is subject to dynamic forces arising from the stochastic and time-dependent spatial distribution of neighbouring particles, themselves undergoing random motion. Third, interactions with the 3D tumour microstructures add additional randomness. This is because the reconstructed microstructure consists of the spaces segmented by tumour cells whose distribution and orientations are random, resulting in variable repulsive and attractive forces and reflections upon collision at the membrane surfaces.

This study focuses on diffusion rather than NP transport in brain tumours, marking other NP transport processes excluded when developing the model. For instance, NPs are assumed to bounce back upon contact with cell membranes, thus excluding cell uptake. However, it has been demonstrated that NP internalisation is also sensitive to surface charge (Patil et al., 2007). To study NP transport in brain tumours, a comprehensive model should be developed to describe the various NP transport mechanisms. In addition to

diffusion, the model should incorporate other mechanisms including NP transport across the tumour microvasculature walls, convection driven by the ISF flow, drug release dynamics, biochemical reactions with proteins and other biomolecules and substances, as well as the cellular uptake of NPs. Particularly, blood vessels in tumours are often tortuous. After intravenous administration, NPs would accumulate around the outer surface of the blood vessel wall. However, these NPs may eventually return to the bloodstream after the administration ends, further decreasing NP concentration within tumour tissue and reducing delivery outcomes. To address this, the model also needs to account for the abnormal tumour microvasculature (Bhandari et al., 2018, 2019; Jain et al., 2007; Zhan et al., 2014a), incorporating vessel features such as heterogeneous distributions, and distinctive vessel morphological characteristics like excessive branching and looping (Bates et al., 2002). This refined model would enable a more comprehensive study of NP transport, rather than just diffusion, in brain tumours.

This study also includes some assumptions and limitations. (1) NPs are assumed to be spherical. In practice, their shapes can differ widely depending on fabrication techniques, NP formulations and materials, loaded drugs, etc. These variations can result in diverse morphologies, including rods, disks, tubes or irregular shapes (Fu et al., 2018; Truong et al., 2015). Consequently, properties such as surface area may differ, which can further influence NP surface charge and thence diffusion. (2) Because of the relatively low velocities of NPs during diffusion in the dense ECM in brain tumours and the strong repulsive interactions between NPs arising from surface charges, collisions between NPs are not taken into account. The modelling framework can be further developed using the Discrete Element Method (Su et al., 2019), which allows for the consideration of both NP collisions and complex NP shapes and morphological features, enabling more NP-specific studies to optimise the design of particular NPs. However, it should also be noted that this approach requires increased computational resources. Additional efforts should be made to enable affordable modelling and computing. (3) The distribution of ECM components can exhibit significant heterogeneity (Andersen et al., 2021), subject to cell types and their activities. Due to the lack of comprehensive data characterising this variability and the small dimension of the computation domain (in 10^{-4} m), a uniform distribution of ECM properties across tumour ECS is assumed in this study. Advanced imaging techniques, such as confocal microscopy for visualising large biomolecules like HA (de la Motte & Drazba, 2011), and second-harmonic imaging microscopy for assessing collagen arrangement (Chen et al., 2012), may provide valuable insight into these biomolecular distributions in tumour ECS. Incorporating these details into the model could relax this assumption. This would be more important when simulating the full transport of NPs in brain tumour tissues.

5. Conclusions

This study numerically assesses NP diffusion in brain tumours of different grades under various conditions, with the accuracy of the employed modelling framework validated against reported experimental data. Our results underscore the critical role of zeta potential in determining NP diffusion. Notably, higher diffusion coefficients can also be achieved by increasing the size of charged NPs while maintaining their zeta potential, owing to the increased total surface charge quantity. An empirical equation is developed to describe the diffusion coefficient as a function of both zeta potential and particle size, providing a direct method to estimate this NP transport property. The positive impact of tissue temperature on NP diffusion is also confirmed. Furthermore, comparisons with the Stokes-Einstein equation reveal the limitation of this equation in directly estimating NP diffusion coefficients in brain tumours. However, this study also demonstrates the potential of the Stokes-Einstein equation as a useful tool for approximating NP diffusion coefficients at different temperatures by scaling values from normal body temperature. Additionally, NP diffusion is slower in the viable regions of high-grade brain tumours despite the higher porosity, largely due to the overexpressed ECM components that increase the resistance. Our modelling results highlight the importance of ECM density in studying NP diffusion in brain tumours, a factor often overlooked in normal tissues. A set of empirical equations is developed to calculate the NP diffusion coefficient in each grade of brain tumour, including the GBM necrotic region, based on HA concentration. By performing pore-scale analysis using the mechanics-based model, this study enhances the understanding of NP diffusion and transport mechanisms in different grades of brain tumours. The findings provide valuable insights for the development of nano-sized drug carriers to achieve deeper tissue penetration and more homogenous drug distribution, ultimately enhancing therapeutic effectiveness.

CRediT authorship contribution statement

Yi Yang: Writing – original draft, Methodology, Visualization, Investigation, Software, Conceptualization. **Ciprian Panaitescu:** Writing – review & editing, Software, Methodology. **Tian Yuan:** Writing – review & editing, Methodology. **Rui Li:** Investigation, Visualization. **Kejian Wu:** Writing – review & editing. **Dubravka Pokrajac:** Supervision, Writing – review & editing. **Yingfang Zhou:** Writing – review & editing, Supervision. **Daniele Dini:** Writing – review & editing, Methodology. **Wenbo Zhan:** Supervision, Conceptualization, Methodology, Writing – review & editing, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Reconstructed tissue microstructures of brain tumours in different grades

The tissue microstructures are reconstructed for brain tumours in all grades based on the geometrical information given in Table 1. The geometries are shown in Fig. A1.

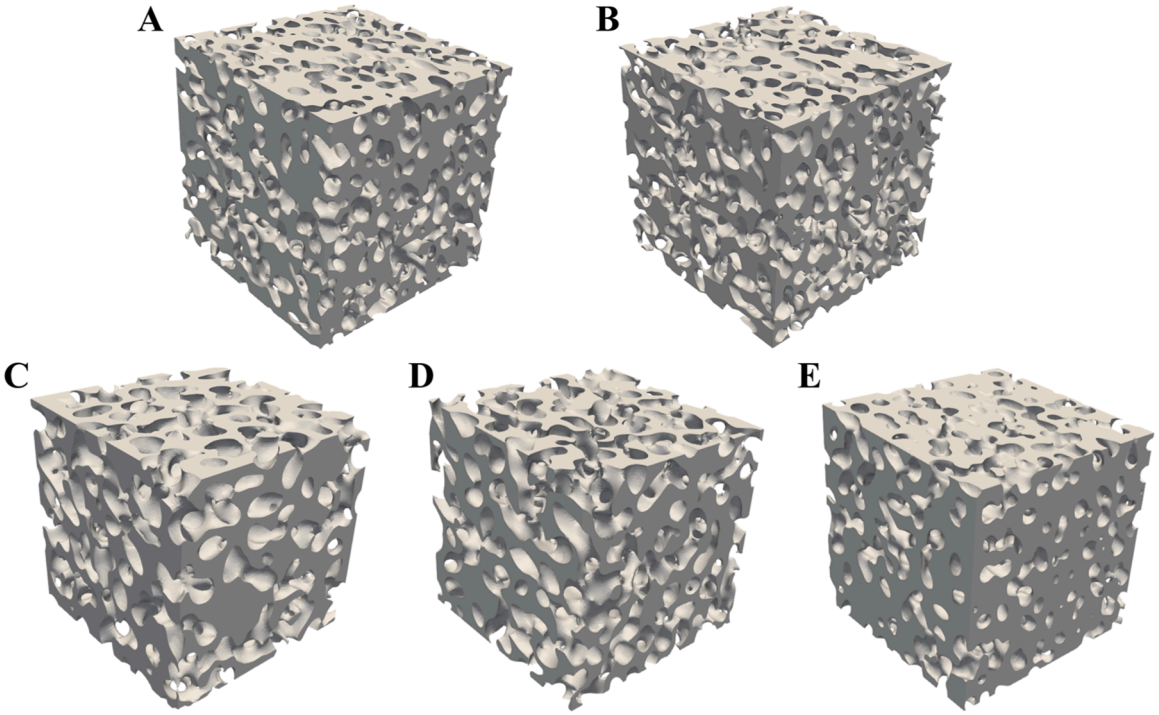


Fig. A1. Reconstructed tumour microstructures. (A) Pilocytic astrocytoma (Grade 1). (B) Diffuse astrocytoma (Grade 2). (C) Anaplastic astrocytoma (Grade 3). (D) Glioblastoma viable region (Grade 4). (E) Glioblastoma necrotic region (Grade 4).

Appendix B. Model solving method for the mathematical model for nanoparticle diffusion

The mathematical model formulated to simulate NP diffusion in brain tumours is solved using the following approach.

1. Mathematical model

Based on the forces acting on an NP listed in Section 2.1, Eq. (1) can be written in the form of the Langevin equation, as

$$m_{\mathrm{p}}\frac{d\boldsymbol{u}_{\mathrm{p}}}{dt} = -\frac{\lambda}{C_{\mathrm{slip}}}\boldsymbol{u}_{\mathrm{p}} + \boldsymbol{F}_{\mathrm{DET}} + \boldsymbol{F}_{\mathrm{B}}(t) \tag{B.1}$$

where $\lambda = 6\pi\mu_{\mathrm{eff}}r_{\mathrm{p}}$ is the damping coefficient. C_{slip} is a slip correction coefficient. $\boldsymbol{F}_{\mathrm{DET}} = \boldsymbol{F}_{\mathrm{C}} + \boldsymbol{F}_{\mathrm{G}} + \boldsymbol{F}_{\mathrm{BY}} + \boldsymbol{F}_{\mathrm{DLVO}}$ is the deterministic force based on the NP size and position. $\boldsymbol{F}_{\mathrm{B}}(t)$ is the random Brownian force (Gaussian white noise).

In this study, the particle relaxation time $t_{\mathrm{r}} = m_{\mathrm{p}}/\lambda$ (Ermak & McCammon, 1978) is orders of magnitude smaller than the simulation time step Δt. For instance, the minimum ratio of $\Delta t/t_r$ in all the simulations is approximately 338. Therefore, the overdamped regime is applicable, as the damping force is significant, and the Langevin equation can be simplified by neglecting the inertial term, $m_{\mathrm{p}}d\boldsymbol{u}_{\mathrm{p}}/dt$ (Carretero et al., 2020; Tuna et al., 2025). This approximation remains consistent with Newton's second law (Contreras-Vergara et al., 2021). Therefore, Eq. (B.1) can be expressed as

$$\frac{\lambda}{C_{\text{slip}}}\frac{d\boldsymbol{r}_{\text{i}}}{dt} = \boldsymbol{F}_{\text{DET}} + \boldsymbol{F}_{\text{B}}(t) \tag{B.2}$$

Eq. (B.2) is multiplied on both sides by C_{slip}/λ to obtain the overdamped Langevin equation, as

$$\frac{d\boldsymbol{r}_{\text{i}}}{dt} = \frac{C_{\text{slip}}}{\lambda}\boldsymbol{F}_{\text{DET}} + \frac{C_{\text{slip}}}{\lambda}\boldsymbol{F}_{\text{B}}(t) \tag{B.3}$$

The Brownian force is satisfied with the fluctuation-dissipation relation and has a Gaussian probability distribution with a correlation function, as

$$\langle \boldsymbol{F}_{\text{B}}(t)\rangle = 0 \tag{B.4}$$

$$\langle \boldsymbol{F}_{\text{B}}(t)\boldsymbol{F}_{\text{B}}(t^{'})\rangle = 2k_{\text{B}}T\lambda\delta(t-t^{'}) \tag{B.5}$$

where the $\delta(t-t^{'})$ function indicates that the force at time t is uncorrelated with the force at any other time (white noise).

Then Eq. (B.3) can be integrated on both sides over the simulation time step Δt from continuous to discrete form, as

$$\Delta\boldsymbol{r}_{\text{i}} = \frac{C_{\text{slip}}}{\lambda}\boldsymbol{F}_{\text{DET}}\Delta t + \frac{C_{\text{slip}}}{\lambda}\int_{t}^{t+\Delta t}\boldsymbol{F}_{\text{B}}(s)\,ds \tag{B.6}$$

Since Brownian force is Gaussian white noise, this integral gives a normal random variable with variance, and the double integral can be given as

$$\left\langle (\Delta W)^2\right\rangle = \left\langle \left(\int_{t}^{t+\Delta t}\boldsymbol{F}_{\text{B}}(s)\,ds\right)^2\right\rangle = \left\langle \int_{t}^{t+\Delta t}ds\int_{t}^{t+\Delta t}ds^{'}\boldsymbol{F}_{\text{B}}(s)\boldsymbol{F}_{\text{B}}(s^{'})\right\rangle \tag{B.7}$$

By using the correlation of the white noise of Eq. (B.5), the double integral can be then expressed as

$$\left\langle (\Delta W)^2\right\rangle = \int_{t}^{t+\Delta t}ds\int_{t}^{t+\Delta t}ds^{'}\,2k_{\text{B}}T\lambda\delta(s-s^{'}) = 2k_{\text{B}}T\lambda\int_{t}^{t+\Delta t}ds^{'} = 2k_{\text{B}}T\lambda\Delta t \tag{B.8}$$

Afterwards, ΔW can be expressed as

$$\Delta W = \int_{t}^{t+\Delta t}\boldsymbol{F}_{\text{B}}(s)\,ds \;\sim\; \mathcal{N}(0,\; 2k_{\text{B}}T\lambda\Delta t) \tag{B.9}$$

The overdamped Langevin equation (Eq. (B.6)) can then be described as

$$\Delta\boldsymbol{r}_{\text{i}} = \frac{C_{\text{slip}}}{\lambda}\boldsymbol{F}_{\text{DET}}\Delta t + C_{\text{slip}}\sqrt{\frac{2k_{\text{B}}T\Delta t}{\lambda}}\cdot\chi = \frac{C_{\text{slip}}}{\lambda}\boldsymbol{F}_{\text{DET}}\Delta t + C_{\text{slip}}\sqrt{2\mathscr{D}\Delta t}\cdot\chi \tag{B.10}$$

where $\mathscr{D} = k_{\text{B}}T/\lambda$. The random displacement at each step is drawn from a normal distribution with variance $2C_{\text{slip}}^2\mathscr{D}\Delta t$, ensuring thermodynamically consistent NP diffusion.

The velocity can be calculated as

$$\boldsymbol{u}_{\text{i}} = \Delta\boldsymbol{r}_{\text{i}}/\Delta t \tag{B.11}$$

Hence, the NP trajectories are updated at each time step in three directions, as

$$x_{t+\Delta t} = x_t + \frac{C_{\text{slip}}}{\lambda}F_{\text{DET,x}}\Delta t + C_{\text{slip}}\sqrt{2\mathscr{D}\Delta t}\cdot\chi_{\text{x}} \tag{B.12}$$

$$y_{t+\Delta t} = y_t + \frac{C_{\text{slip}}}{\lambda}F_{\text{DET,y}}\Delta t + C_{\text{slip}}\sqrt{2\mathscr{D}\Delta t}\cdot\chi_{\text{y}} \tag{B.13}$$

$$z_{t+\Delta t} = z_t + \frac{C_{\text{slip}}}{\lambda}F_{\text{DET,z}}\Delta t + C_{\text{slip}}\sqrt{2\mathscr{D}\Delta t}\cdot\chi_{\text{z}} \tag{B.14}$$

where $\chi_x,\ \chi_y,\chi_z\ \sim\ \mathcal{N}(0,1)$.

This final form provides the foundation for the numerical integration of NP trajectories induced by deterministic force and Brownian motion.

2. Discretisation and time integration

Since both deterministic and stochastic forces are included in our modelling study, the deterministic update and stochastic update of NP trajectories are achieved by the explicit Euler scheme and the Euler–Maruyama method for the numerical solution of a stochastic differential equation, respectively.

$$\Delta \boldsymbol{r}_{\mathrm{i,DET}} = \frac{C_{\mathrm{slip}}}{\lambda}\boldsymbol{F}_{\mathrm{DET}}\Delta t \tag{B.15}$$

$$\Delta \boldsymbol{r}_{\mathrm{i,\ Brownian}} \sim \mathcal{N}\left(0, 2C_{\mathrm{slip}}^2\mathscr{D}\Delta t\right) \tag{B.16}$$

3. Initial condition

1000 particles are initialised and released, with zero velocity, to random locations within a sphere volume at the centre of the tumour ECS microstructure domain, as

$$\boldsymbol{u}_{\mathrm{p,i}}(t=0) = 0, \forall \mathrm{i} \in \{1,\ 2,\ 3,\ \ldots,\ 1000\} \tag{B.17}$$

4. Boundary conditions

If at the end of a time step the calculated position of an NP is within a cell, the NP is assumed to have bounced back from the cell membrane and continued diffusing. This is described as

$$\boldsymbol{r}_{\mathrm{i}}(t+\Delta \mathrm{t}) = \boldsymbol{r}_{\mathrm{i}}(t) \tag{B.18}$$

$$\boldsymbol{u}_{\mathrm{p,i}}(t+\Delta \mathrm{t}) = -\boldsymbol{u}_{\mathrm{p,i}}(t+\Delta \mathrm{t}) \tag{B.19}$$

If an NP diffuses outside the simulation domain, it is permanently excluded from the current simulation.

Appendix C. Complementary details of the program

A Python program is developed to implement the model using several libraries and packages, including the Python standard library, numerical Python (numpy), pandas, matplotlib pyplot interface, scipy, scikit-image, and pyvista.

Fig. C1 presents the flowchart outlining the key steps and logic of the program, with a detailed description provided below.

1. Initialisation:

Different types of inputs are imported into the model, including

- Model geometry: 3D tumour microstructure
- Model parameters: NP properties (particle size, zeta potential), tumour property (HA concentration), environmental factor (temperature), and parameters in Table 2
- Solver setting: total number of NPs, initial NP-packed region, time step, simulation time

The Euclidean Distance Transform (EDT) (Maurer et al., 2003) calculates the shortest distance between each NP and the nearest cell membrane at the initial time and after each time step. Additionally, the gradients of the EDT in x, y, and z directions are also computed, which are used to determine the directional components of the force that cell membranes exert on each NP, $\boldsymbol{F}_{\mathrm{DLVO}}$. This force is then used to calculate the deterministic force, $\boldsymbol{F}_{\mathrm{DET}}$, acting on each NP directly (Eq. (B.1)).

NPs are randomly located in the initial NP-packed region with zero initial velocity. The initial position of each NP is saved. The time-course plot of *MSD* is initialised and will be updated during the simulation.

2. Simulation loop (pseudocode):

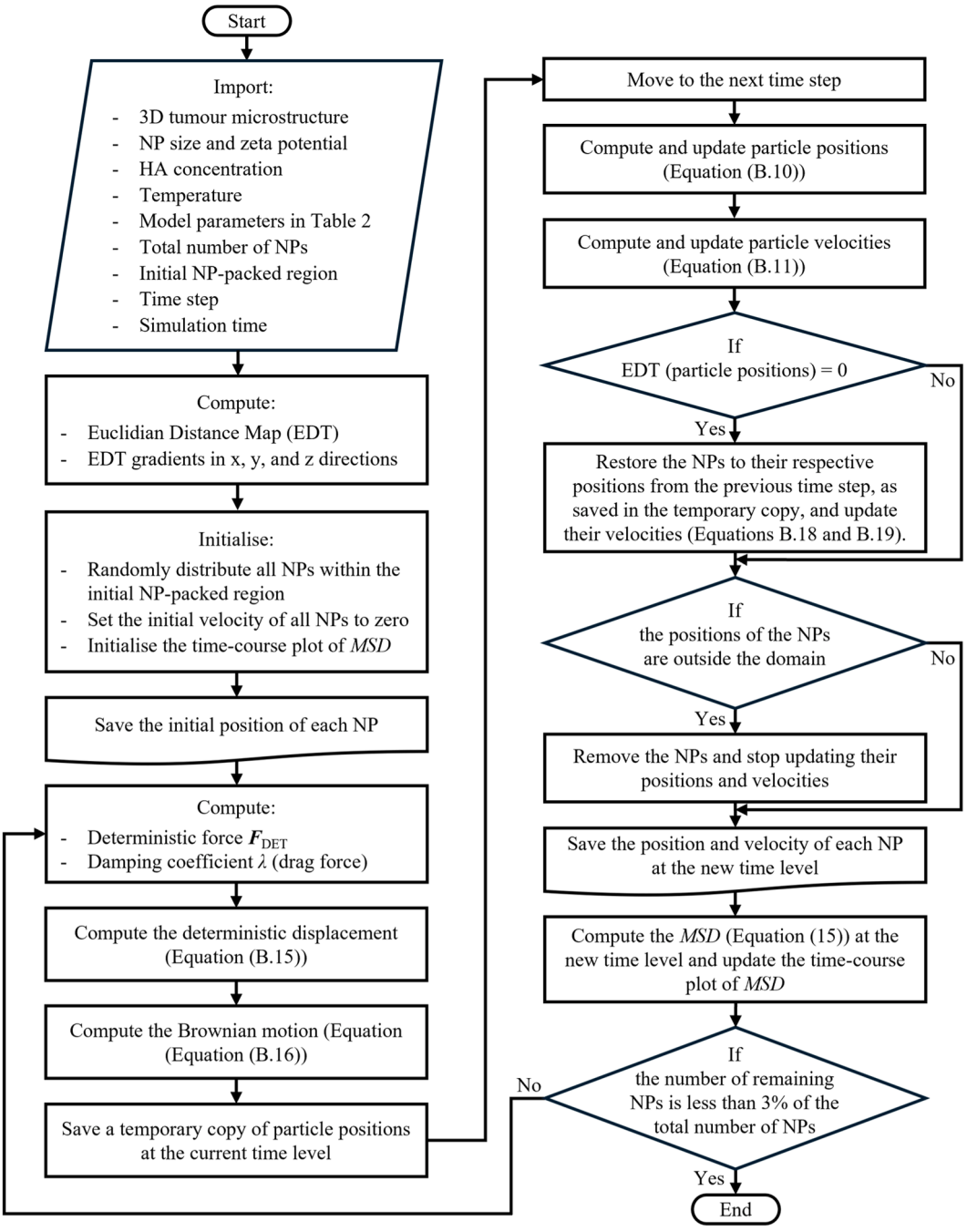


Fig. C1. Workflow of the program built in Python for NP diffusion in brain tumours.

Appendix D. Modelling strategy at the pore-scale

The modelling strategy employed in this pore-scale study is schematically represented in Fig. D1. The developed particle-tracing model and reconstructed 3D tumour microstructure are used to simulate NP motion within the tumour ECS, accounting for interactions between NPs and between NPs and the tumour microenvironment. NP motion is quantified by tracking individual velocities and positions as they diffuse from the centre of the ECS, where the NPs are released, towards the outer surfaces of the ECS domain. The resulting pore-scale trajectories are used to compute the *MSD*, which is subsequently used to derive the diffusion coefficient of NPs at the tissue (macro) scale. This macroscale transport property captures the combined effects of NP intrinsic properties and the tumour microenvironment. The approach provides a mechanistic link between microscale NP dynamics and macroscale transport phenomena.

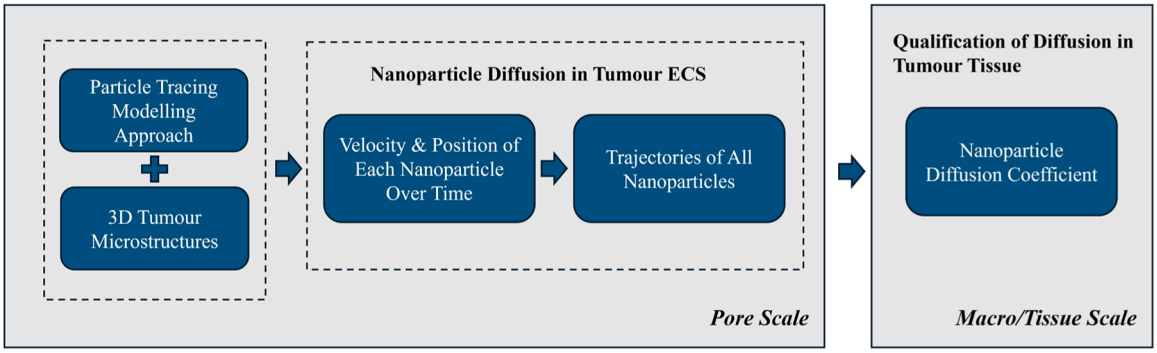


Fig. D1. Schematic diagram illustrating the particle tracing modelling approach.

Data availability

The datasets generated and/or analysed during the current study are available from the corresponding authors on reasonable request.

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